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Integrated planning of net-zero power systems for all

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Achieving global net-zero power systems by mid-century demands integrated frameworks addressing climate mitigation and energy access equity. Here we present a spatio-temporally resolved global power system model (0.25° × 0.25°, 8,760 hours) co-optimizing capacity expansion and operational strategies. Findings show that net-zero global power systems meeting universal electricity needs for decent living standards are technically feasible, requiring 15–20 TW of variable renewable energy (VRE). Abundant VRE resources offer cost-effective electricity access in low-income regions, such as Africa, promoting climate justice. Land use is critical, with solar photovoltaics alone requiring over 9 million hectares. Over 80% of VRE is within 200 km of load centres. Demand-side management could reduce system costs by 6.5% (~US\$182 billion yr^{-1}). Expanding international transmission and removing renewable technology trade barriers could cut costs by 5.6% (~US\$157 billion yr^{-1}) and 12.2% (~US\$345 billion yr^{-1}), underscoring the pivotal role of international collaboration in building inclusive net-zero power systems.

Energy itself is not what people want, but the services provided by energy and the products relying on these services are[1]. As the most pervasive energy carrier, electricity has become increasingly critical for modern economies. Global electricity consumption has surged more than fivefold over the past five decades to about 30 PWh yr^{-1} (refs. 2,3). This rapid expansion of electricity use creates tension between sustainable development imperatives and growing electricity demand, which is accelerated by emerging technologies such as artificial intelligence[4]. First, the electricity sector's contribution to anthropogenic climate change represents a primary challenge. Since 1974, cumulative carbon emissions from electricity generation have reached ~500 gigatonnes (Gt) (ref. 5), accounting for 35–40% of total anthropogenic emissions during this period. To align with international climate targets, future emissions from the electricity sector must decrease dramatically, achieving net- zero or negative emissions by mid-century, while accommodating expanded demand for end-use electrification in hard-to-abate sectors[6–8]. Second, energy inequalities persist globally:although average world electricity consumption per capita now has reached over 3,000 kWh yr^{-1} (refs. 9,10), more than one billion people lack reliable electricity access, with sub-Saharan African households consuming merely 300 kWh annually on average[11]. These dual imperatives—deep decarbonization and equitable energy

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access—present humanity with an unprecedented challenge: transforming the electricity sector as industrial society's principal energy carrier in a manner that addresses both climate constraints and distributive justice requirements[12,13].

This climate–equity nexus aligns directly with the Paris Agreement's 1.5–2 °C guard-rail[8] and the United Nations' Sustainable Development Goal[14], a mission hinging on two sectoral advantages. One is technologically viable benefiting from variable renewable energy (VRE), energy storage and grid innovations; the other is geopolitically equitable as renewable resources (wind and solar) are universally accessible, exceeding national demand thresholds across all regions[15,16]. Recent energy transitions have demonstrated remarkable progress, with VRE technologies—particularly wind and solar photovoltaic (PV)—undergoing exponential growth. Global installed capacity surpassed 2,000 GW in 2024[17], establishing VRE as a cornerstone to replace fossil fuels in the low-carbon power system. Meanwhile, the average levelized cost of utility-scale solar PV declined by 90% between 2010 and 2023, from US\$0.460 to US\$0.044 kWh^{-1} (ref. 18). This dramatic cost reduction enhances VRE's potential to address climate mitigation and reinforce the nexus by enabling the electricity access required for decent living standards (DLS)[13,19,20]. However, increasing penetration of VRE also poses both technical and social equity challenges. First, the inherent intermittency and variability of VRE generation present substantial challenges to integration and exacerbate system volatility, thereby necessitating major investment in ancillary services and supporting infrastructures[21–23]. Additionally, large-scale VRE deployment may result in unprecedented land-use requirements[23–26], competing with agricultural, residential and conservation priorities for limited resources[25]. Second, social equity challenges would emerge because insufficient planning of VRE can exacerbate energy inequalities through disproportionate burden distribution or limited access to energy transition opportunities[27]. Therefore, it necessitates integrated planning of power systems that maintain grid reliability while incorporating spatially resolved optimization of land-use constraints, resource availability and distributional equity for large-scale VRE expansion[24,28–31].

Analyses of global transitions towards net-zero electricity systems have long depended on integrated assessment models (IAMs). However, their structural limitations in capturing the dynamics of electricity systems with high-share VRE technologies may lead to a misinterpretation of the challenges and opportunities inherent in the unprecedented energy transition[32–34]. The characteristic spatio-temporal coarseness of IAMs—typically employing annual temporal resolutions and continental-scale regional aggregations—fails to resolve critical operational constraints such as renewable intermittency, sub-seasonal storage demands, ramping limits of thermal and nuclear power, hydropower reservoir management and cross-border/inter-grid transmission dependencies. Moreover, the technological simplifications in IAMs, through aggregated generation categories and exclusion of grid-integration parameters, obscure location-specific infrastructure trade-offs[35,36]. These shortcomings hinder their ability to demonstrate (1) technology configurations ensuring supply–demand equilibrium and emissions constraints, (2) wind and solar spatial deployment layout that can balance land-use constraints and integration costs and (3) investment planning aligned with financing readiness[33–35,37].

Recognizing these limitations, some recent studies focusing on global power system transitions have made progress in incorporating hourly balancing of supply and demand. However, as for the capacity expansion, they either apply heuristic optimization methods where VRE siting is determined by minimizing local LCOE[38,39], a metric known to be insufficient as it ignores the declining market value of VRE at high penetrations[40,41] or rely on a simplified power system expansion model (PSEM) with insufficient temporal coverage (tens or hundreds of hours) and VRE spatial resolution (power grid or sub-region level)[42] or even directly adopt capacity results from IAMs[43] that are prone to aforementioned weaknesses. None of them has achieved full-scale capacity expansion optimization considering hourly operation across 8,760 hours in a full year simultaneously. Given that capacity deployment decisions feature future power systems and previous works have demonstrated that insufficient resolution and model framework may result in notable bias in capacity expansion and system cost estimates[23,44], navigating complex interdependencies inherent in global power system decarbonization demands a next generation of modelling frameworks. These must, in a sophisticated manner, integrate hourly grid operations, spatially explicit optimization for VRE, hydropower resources and carbon storage and international transmission network planning. Advancing this modelling and computational frontier—essential for mapping equitable decarbonization trajectories—requires an unprecedented synergy between geospatial analytics, power system engineering and energy economics, so that the model would be able to address several interrelated critical questions: how is the power system that realizes both net zero and universal access structured? What would be the global geographical distribution of technology adoption? And how would international collaboration bring down the costs of such net-zero power systems that serve all people?

In this study, we develop the Global Integrated Sustainable Power-system Optimization (GISPO) model, a global PSEM with hourly temporal resolution (8,760 hours annually) and 0.25° × 0.25° geospatial optimization granularity for VRE. The GISPO model is designed to explore the viable patterns of net-zero power systems by mid-century (2050) while addressing the resource constraints and just transition issues. This integrated model co-optimizes capacity expansion and operational strategies across multiple energy technologies: wind (onshore and offshore), solar PV (utility-scale and distributed), hydropower (reservoir and run of river), nuclear, thermal plants (carbon capture-equipped or conventional coal, gas and biomass), transmission infrastructure (intra-grid, inter-grid and cross-border), storage technologies (pumped hydro storage and battery) and negative emissions technologies (NETs, biomass energy or direct air capture with carbon storage), while ensuring net-zero CO_2 emissions compliance and meeting electricity demand thresholds for decent living standards. The GISPO employs a geospatial architecture that divides the world into 91 regions and 144 grids according to grid operation[45], with finer granularity for large nations such as China and the USA, where the electricity demand and transmission are balanced on each grid. This model advances existing energy modelling through several transformative features: (1) it is a temporally and spatially resolved power system expansion model to co-optimize capacity expansion for each generation technology and transmission line and hourly system operation (8,760 hours) for a planning year at the global scale; (2) it enables grid-cell level (approximately 600,000 cells at 0.25° × 0.25° in total) capacity expansion for VRE, facilitating precise siting and generation complementarity analyses between wind and solar resources; (3) it incorporates dam-site-level hydropower planning with reservoir management and (4) it endogenously matches carbon sources and sinks through integrated carbon capture and storage (CCS) network optimization. Leveraging these features, we also assess the system's sensitivity to alternative assumptions about energy equity, technology cost trajectories, inter-grid transmission capacity, renewable resource availability and international trade dynamics under multiple scenarios. By synthesizing geospatial, technological and socio-economic parameters, the GISPO elucidates crucial challenges and strategic opportunities inherent in the pursuit of net-zero power systems for all regions.

Net-zero power systems for decent living

To address energy justice for underserved populations while forecasting future electricity demand, this study moves beyond reliance on historical trends by developing a two-step framework for projecting country-level electricity demand to 2050. First, we establish a

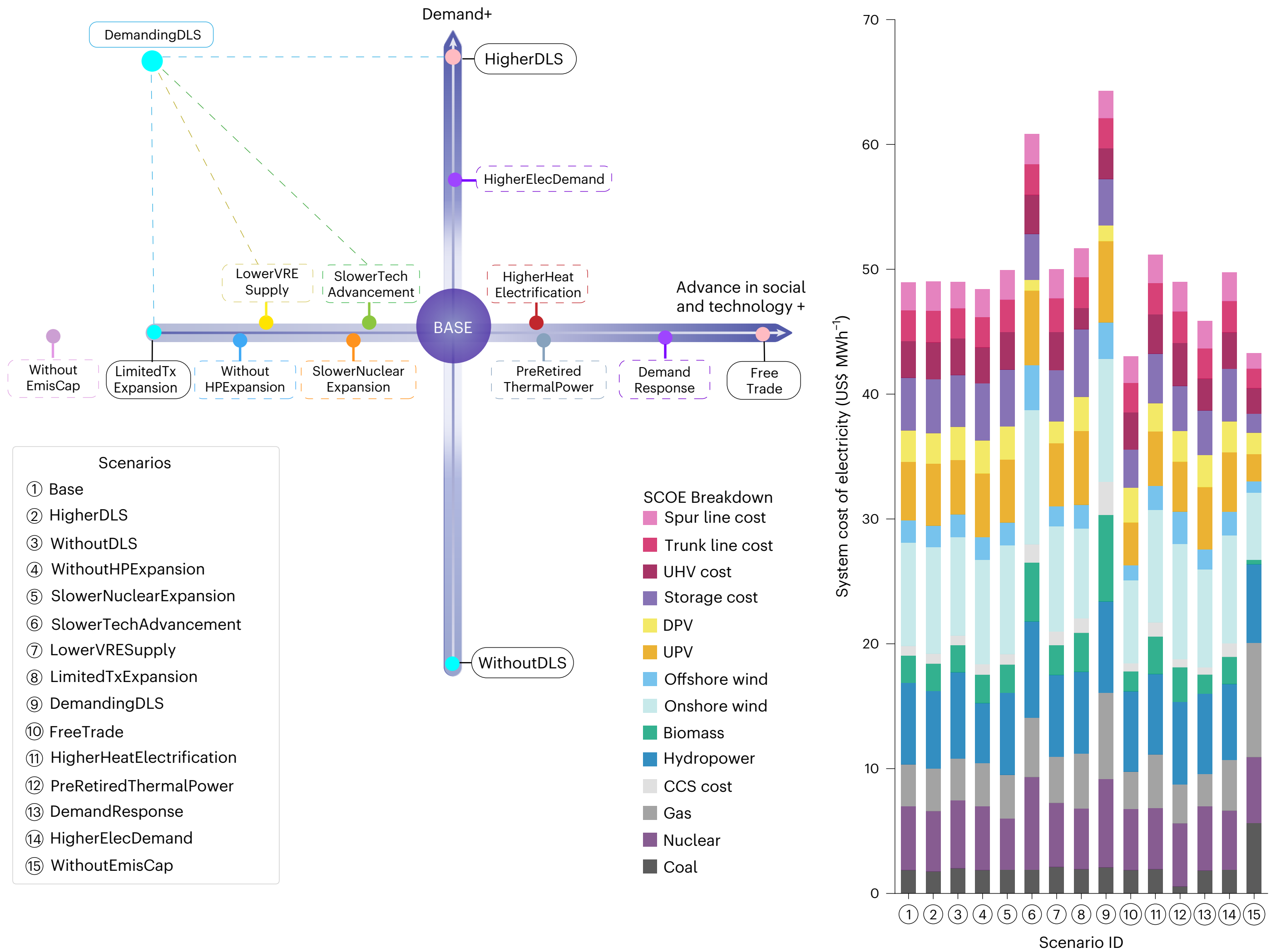

Fig. 1 | Scenario framework and associated SCOE for net-zero power systems. The left panel illustrates the conceptual design of 15 scenarios, diverging from the BASE scenario along axes of demand growth and socio-technological advancement (scenario definition in Extended Data Table 1). The right panel presents the corresponding SCOE (US dollar per megawatt-hour) breakdown by technology and transmission infrastructure for each scenario. SCOE is defined as the annualized capital and operational expenditures divided by total electricity demand, excluding distribution and administrative costs. UHV: ultra-high voltage; DPV: distributed photovoltaic; UPV: utility-scale photovoltaic; CCS: carbon capture and storage.

normative baseline using United Nations medium-variant population projections[10] and a globally recognized minimum final energy threshold for decent living standards (15 GJ yr^{-1} per capita[46]). Assuming a 60% electrification rate[47], this translates to an electricity-specific demand of ~2,500 kWh yr^{-1} per capita, thereby embedding energy access equity. We then compare these normative projections with forecasts derived from the extrapolation of historical consumption data and select the higher of these two values as the projected demand for each country. The base scenario yields a global electricity demand of approximately 58.3 PWh yr^{-1} by 2050 (about twice the current level[3]) for the GISPO model. Our multiple-scenario analyses demonstrate that achieving a global net-zero power system to meet this projected demand for all by 2050 is not only technically feasible but also economically viable (scenario definition and key findings across scenarios are available in Extended Data Table 1). This finding exhibits robustness against stringent assumptions regarding equity, land use, transmission infrastructure and technological advancement.

Figure 1 illustrates the scenario matrix and the system cost of electricity (SCOE)—defined as annualized capital and operational expenditures divided by total demand, excluding distribution and administrative costs[28]. Our base scenario analysis projects a considerable reduction in SCOE, from approximately US\$76 MWh^{-1} in 2022 (derived from operational optimization using the GISPO model with fixed installed capacity) to US\$49 MWh^{-1} in the 2050 net-zero system. As shown in Fig. 2, this future system is characterized by a dominant VRE penetration rate of 70%, anchored by substantial capacities: 5,588 GW onshore wind (generating 19.3 PWh yr^{-1}), 768 GW offshore wind (3.4 PWh yr^{-1}), 9,082 GW utility-scale solar PV (UPV, 13.9 PWh yr^{-1}) and 4,306 GW distributed solar PV (DPV, 6.0 PWh yr^{-1}). Nuclear power emerges as the second-largest carbon-free source, expanding to 1,335 GW capacity and contributing 10.4 PWh yr^{-1}, with the grid-level installation upper bound is downscaled from ref. 48 (Supplementary Section 3.4.2). High-resolution resource and dam-site modelling with conservation constraints indicates 1,643 GW of hydropower installed capacity, providing 5.7 PWh annually (approximately 9% of total demand). Consistent with assumed end-of-life retirement or CCS retrofitting, legacy thermal capacity consists of 1,145 GW coal (30 GW with CCS) and 2,063 GW gas (10 GW with CCS), collectively supplying 2.9% of total demand, with combined heat and power (CHP) plants operating as necessary in winter. Biomass systems scale to 166 GW (only 4% of technical installation potential assessed in this study) with generation of 972 TWh yr^{-1} (less than 10 EJ yr^{-1} fuel combusted), of

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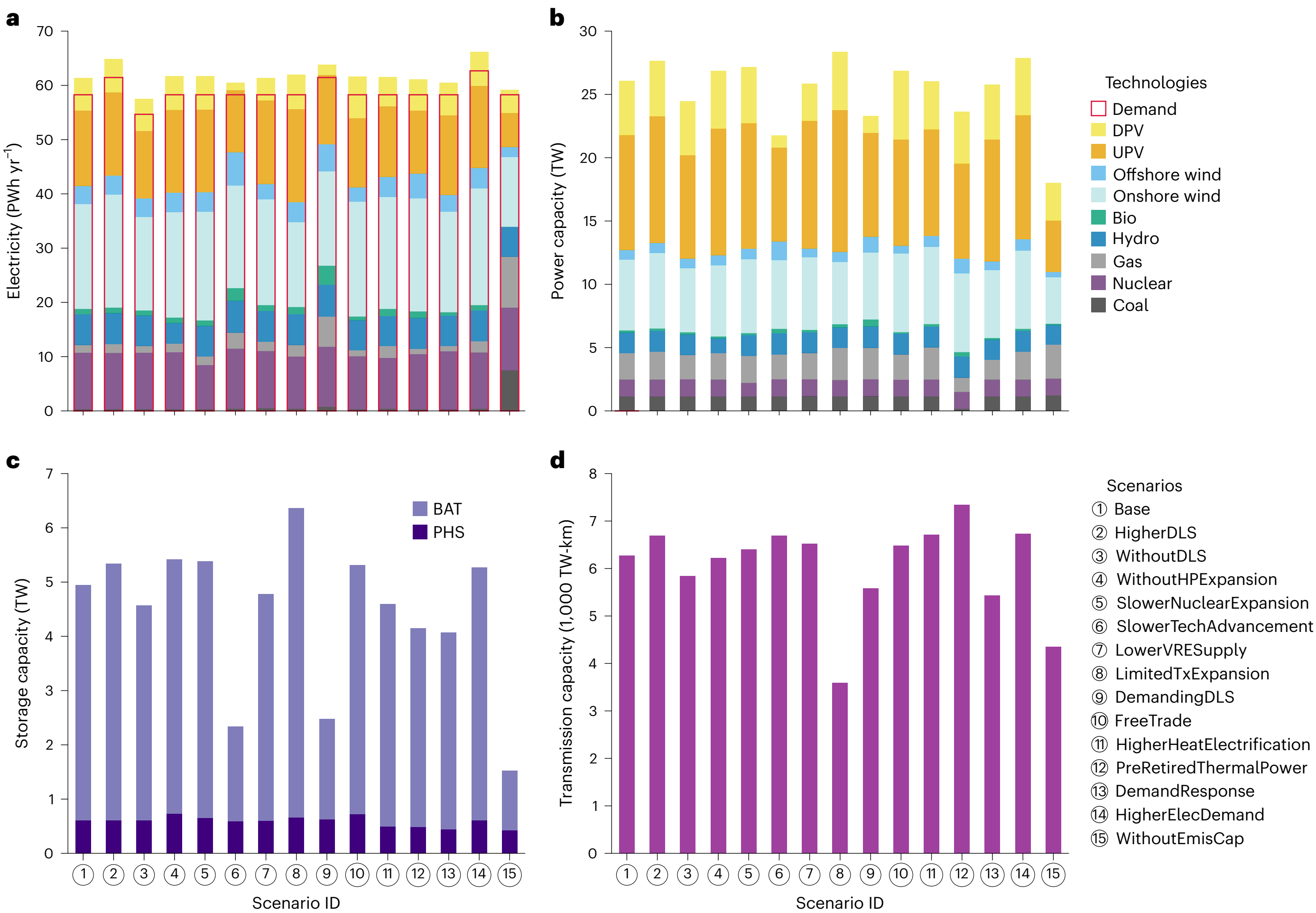

Fig. 2 | Key features of optimized net-zero power systems across scenarios. **a**, Generation mix (petawatt-hour per year, DPV: distributed photovoltaic; UPV: utility-scale photovoltaic). **b**, Capacity mix (terawatt). **c**, Installed storage capacity (terawatt) of pumped hydro storage (PHS) and batteries (BAT). **d**, Transmission capacity (1,000 TW-km). Within the planning year (2050), installed coal and gas power capacities by 2022 will undergo retirement upon reaching their designed service life or will be subject to retrofitting with carbon capture equipment as necessitated by the optimization. Unless otherwise specified (for example, the WithoutEmisCap scenario), the annual emissions target for all scenarios in this study is net zero. Subsidies for early retirement of coal-fired and gas-fired power plants are not incorporated into the system cost in the PreRetiredThermalPower scenario. Scenario definitions are provided in Extended Data Table 1. See regional/power-grid-level power and generation mix in Supplementary Figs. 85–114.

which 107 GW installations are biomass energy with carbon capture and storage (BECCS). Combined with 11.3 Mt CO_2 yr^{-1} of direct air capture with carbon storage (DACCS) deployment, the power systems achieve 910 Mt CO_2 yr^{-1} in net-negative emissions. Successful integration of this high VRE share relies on balancing infrastructure, including 22.2 TWh of total storage (607 GW pumped hydro storage, 4,336 GW battery) and an inter-grid 6,300 TW-km transmission network, ensuring global wind and solar PV curtailment below 5%. Realizing this sustainable power system (comprising VRE and storage) also necessitates substantial materials supply (Supplementary Fig. 58 provides estimates).

Analysis of alternative scenarios reveals that, while varying energy justice and electricity demand assumptions impact the optimized net-zero power system's structure (Fig. 2), the overall SCOE remains remarkably stable. We examined a HigherDLS scenario, increasing the per capita electricity floor for DLS by 40% to 3,500 kWh yr^{-1} (61.5 PWh yr^{-1} total demand), and a WithoutDLS scenario, omitting decent living standards from demand projections and only considering historical trends. The HigherDLS scenario requires a 5.5% installed capacity increment, met largely by more installation of utility-scale solar PV (+ 10.8%) and onshore wind (+7.8%), supported by expanded storage (+7.2%) and inter-grid transmission (+6.7%). Conversely, the WithoutDLS scenario reduces total generation by 6.4% but disproportionately cuts decentralized solutions, notably distributed solar PV (−28.7%) and battery storage (−9.3%), impacting electricity access in underserved regions. Despite these substantial structural adjustments, the SCOE deviates by less than 1% from the base scenario's US\$49 MWh^{-1}. We examine an even higher electricity demand scenario (HigherElecDemand) that maintains the lower bound of DLS energy requirements (15 GJ yr^{-1} per capita) while assuming elevated electrification rates (72%) and ensuring that each country's 2050 demand reaches at least 1.5 times current levels, which yields a global electricity demand of 62.7 PWh yr^{-1} by 2050. Meeting this increased demand requires proportionally greater installed capacity—solar PV expands by +7%, wind by +11.2% and energy storage by +6% relative to the base scenario—yet the SCOE rises by less than 0.1%. This cost resilience demonstrates the capacity of renewable-dominated systems, complemented by storage and transmission infrastructure, to cost-effectively absorb variations in demand levels and spatial distribution through optimized resource deployment. These findings indicate that prioritizing energy justice can be achieved with minimal impact on overall system cost.

The results across multiple scenarios consistently demonstrate the inherent feasibility of achieving equitable net-zero power systems, even when subjected to conservative assumptions regarding technological transitions. The SlowerTechAdvancement scenario, defined by a simulated delay in cost reductions for renewable and storage technologies, imposes the most substantial economic burden, elevating the SCOE by 24% to US$61 MWh^{-1}. This is driven by a reduced deployment of utility-scale PV capacity (7,398 GW versus 9,082 GW in the base scenario) and a threefold increase in the reliance on fossil generation with CCS. Nevertheless, net zero is preserved through a substantial expansion of BECCS, exceeding the base scenario capacity by 30%. In a more conservative VRE supply assumption, the LowerVRESupply scenario, which restricts the land availability for siting VRE, results in a marginal 2% SCOE increment to US$50 MWh^{-1}. The climate-equity mission is primarily achieved through strategic increases in bioenergy capacity (+7% to 178 GW, contributing an 8% rise in generation) and utilization of energy storage. The LimitedTxExpansion scenario, imposing limits on the expansion of cross-border transmission, can also successfully achieve emissions targets and equitable access by leveraging regional offshore wind deployment (+92% capacity) and enhancing demand-side flexibility alongside increased storage (+27% to 28 TWh), although leading to a 5.6% rise in system costs (approximately US$157 billion annually). Notably, the DemandingDLS synthesis scenario, which integrates all the above-analysed constraints and incorporates stringent energy justice objectives, underscores the system's profound resilience under extreme conditions. While incurring a substantial 31% cost increase (US$65 MWh^{-1}), it ensures universal access through highly localized renewable deployment and aggressive scaling of BECCS and DACCS (3.5 times the base capacity), thereby achieving net zero without sacrificing minimum consumption standards.

The intra-annual variability of renewable energy (Supplementary Fig. 7) necessitates high temporal coverage in power system modelling, challenging traditional typical-hour approaches[28].

To quantify this impact, we design 20 capacity expansion experiments using reduced time horizons (12 days to 6 months; Supplementary Section 7) against a full 8,760-hour base scenario. We find that reduced-coverage models underestimate global renewable and storage capacity (Extended Data Fig. 1), though results vary across different regions (Supplementary Figs. 152–156). Quantitatively, these scenarios reduce onshore wind and utility-scale solar capacity by up to 11.5% and 20%, respectively, compared to the base scenario. System flexibility assets show even greater sensitivity: offshore wind capacity fluctuates dramatically (−20% to +25%), while required battery storage varies (−14% to +6%) and pumped hydro exhibits the largest variance (−29% to +13%), demonstrating that non-8,760 models fail to capture critical system needs.

Spatial siting and land-use requirement

Figure 3 illustrates the optimized spatial distribution of wind and solar PV capacities at a 0.25° × 0.25° grid resolution in the base scenario. This distribution incorporates both existing installations (on the assumption that they will be renewed before 2050) and new capacity additions. The deployment of global wind power, encompassing both onshore and offshore installations, exhibits important geographical concentration. Over 55% of the total wind capacity is projected to be located in five nations: China (2,017 GW), the USA (761 GW), Russia (395 GW), India (283 GW) and Canada (218 GW). Several additional jurisdictions are projected to exceed 100 GW of installed wind capacity, including West Africa (175 GW), North Africa (168 GW), Vietnam (168 GW), Sudan–Eritrea (167 GW), Brazil (154 GW), Egypt (119 GW), Morocco (129 GW) and the British Isles (112 GW). Collectively, Europe is estimated to require 1,090 GW of wind installations. In Africa, onshore wind emerges as the predominant electricity source, contributing over 4.5 PWh yr^{-1} from 1,138 GW of installed capacity. Offshore wind presents opportunities as a clean energy source for land-constrained or high-latitude regions and as a potential component of marine integrated energy hubs, owing to its high capacity factors. Our analysis projects a global offshore wind demand of 768 GW by 2050. China's substantial deployment of 264 GW, with 95% concentrated along its high-demand, land-scarce eastern and southern coasts, dominates this sector. Other critical offshore development zones include Vietnam (162 GW), the European North Sea (120 GW), the USA (37 GW) and Canada (21 GW), underscoring offshore wind's crucial role in sustainable energy transitions. Spatial siting of VRE across scenarios is listed in Supplementary Figs. 61–75.

Our analysis suggests that deploying over 13,000 GW of solar PV capacity could generate approximately 20 PWh yr^{-1} of clean electricity by mid-century. Spatial optimization results from the GISPO model reveal that nearly 50% of this capacity is concentrated in China (3,336 GW, yielding 4.5 PWh yr^{-1}) and India (3,127 GW, yielding 4.4 PWh yr^{-1})—the two most populous nations projected for 2050. Although these two countries exhibit similar installed capacities and generation output, China's lower curtailment rate (3%) compared to India's (8%) is attributed to its cost advancements in energy storage technologies and more flexible inter-grid and cross-border UHV transmission infrastructure (that is, with higher regulation capability). Furthermore, several other nations are projected to exceed 200 GW of installed solar capacity, including the USA (756 GW), Saudi Arabia (554 GW), Pakistan (532 GW), Indonesia (491 GW), Thailand (241 GW), Japan (225 GW) and Vietnam (202 GW), highlighting the important potential for solar deployment even in densely populated regions. At the continental level, Asia is projected to dominate solar PV installations with 10,385 GW, followed by Africa (1,179 GW), North America (923 GW), South America (422 GW) and Europe (400 GW).

Distributed solar PV (DPV) is anticipated to play an integral role in low-carbon and equitable power systems, offering advantages in land conservation and deployment flexibility. In this Article, we develop a high spatial resolution framework for estimating building footprint areas utilizing a data-driven machine learning approach that integrates high-resolution satellite imagery with point-of-interest geographic data[49,50], with details shown in Supplementary Section 2.1.4. Factors such as public acceptance and rooftop orientation are considered using a rule-of-thumb availability factor (available rooftop area to total area) of 0.35 for resource assessment[50]. This framework reveals a global DPV installation potential exceeding 10,000 GW and a generation potential exceeding 16.7 PWh yr^{-1}. Within the GISPO model, which assumes zero grid-integration costs and higher investment costs for distributed solar PV (for higher levelized cost and local integration in the real world) than utility-scale solar PV, the optimization results indicate a requirement for 4,360 GW of DPV by 2050 to provide approximately 6.0 PWh yr^{-1} of electricity. Large-scale DPV deployments are expected to be predominantly located in China (1,484 GW), India (856 GW), Indonesia (227 GW), Thailand (164 GW), Pakistan (135 GW), the USA (118 GW), Brazil (116 GW) and Vietnam (108 GW). Notably, Brazil's DPV build-out is expected to substantially surpass its utility-scale solar PV capacity (18 GW). This disparity is driven by stringent environmental protection measures, such as safeguarding tropical rainforests and elevated grid-integration costs that enhance the economic attractiveness of DPV.

The spatial deployment of renewable energy infrastructure presents an important implication: grid-integration costs, which are endogenously considered for new VRE expansion, fundamentally shape the installation patterns of solar PV, often prioritizing proximity to load centres over resource abundance. Existing installations clearly exemplify this counterintuitive trend. Xizang, despite possessing China's most favourable solar radiation resources[15], reported substantially lower installed solar PV capacity in 2022 compared to China's eastern provinces, which have much more concentrated electricity consumption but lower solar resource potential. While this phenomenon has been documented in regional studies[23,24], its broader implications for global-scale renewable energy deployment have remained largely unexplored. To fill this gap, the GISPO model explicitly incorporates two critical transmission components for wind, solar PV and hydropower: spur lines connecting generation facilities from cells/dam sites to substations and trunk lines linking substations to load centres[28]. Both sets of distances are optimized to minimize total length. Drawing

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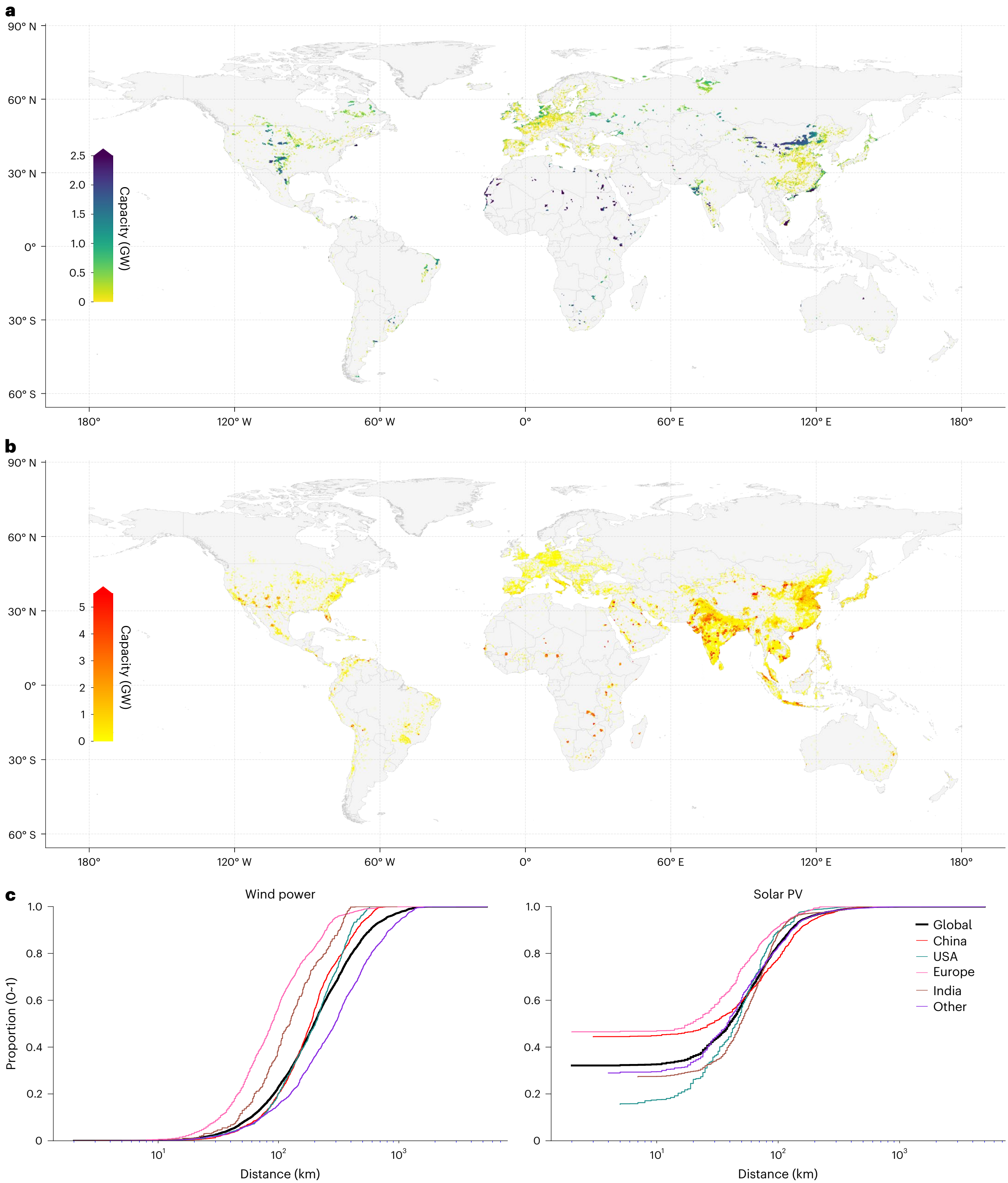

Fig. 3 | Optimized deployment of variable renewable energy. a,**b**, Cell-level installation (gigawatt) of wind (onshore + offshore) (**a**) and solar photovoltaic (utility-scale + distributed) (**b**). **c**, Cumulative distribution of total distance (kilometre) of spur and trunk lines that connect cells (weighted by planned capacity) to their corresponding load centre. Basemap data in **a** and **b** from Natural Earth (https://www.naturalearthdata.com/).

upon high-resolution substation and load-centre data derived from OpenStreetMap and satellite-identified urban areas (detailed in Supplementary Section 3.5.1), our spatially resolved modelling framework indicates that these spur and trunk line costs constitute 9.6% of total system costs (Fig. 1), a factor often overlooked in many energy models[38]. These substantial grid-integration costs necessitate revisions in development potential assessments for resource-rich regions. In the base scenario, Xizang's projected solar PV deployment is only 92 GW, representing less than 3% of China's total solar capacity. Similarly, the Sahara Desert region, while possessing both superior solar resources and vast available land, is projected to deploy less than 300 GW of solar PV capacity—merely 2.5% of global installations. This is the case even when accounting for flexible transmission line expansion within Saharan countries and interconnections to Europe and Asia via the Middle East. However, under the HigherDLS scenario, where local power demand increases for improved living standards, we observe a 130% increase in solar PV installed capacity (exceeding 700 GW) in countries around the Sahara Desert. This demonstrates that integrating renewables for local demand surpasses long-distance transmission in feasibility, particularly for large-scale Saharan renewable energy initiatives aimed at environmental improvement[51]. These findings are further supported by our capacity-weighted distribution analysis (Fig. 3c), which reveals that, globally, over 50% of installed wind capacity and more than 80% of installed solar PV capacity are located within 200 km and 100 km of load centres, respectively, and corroborating similar findings previously for China[24]. Cumulatively, over 80% of the total installed VRE capacity (wind + solar PV) is positioned within a radius of approximately 200 km from load centres (Supplementary Fig. 76). Although regional variations are observed, they generally conform to these overarching global patterns, as depicted in Fig. 3c. For instance, in Europe, which possesses a more extensive and distributed network of load centres, over 55% of wind and 90% of solar PV capacity are situated within 100 km of load centres.

The projected deployment of over 20,000 GW of VRE globally by 2050 calls for careful consideration of land-use implications, particularly for solar PV installations. This concern is underscored by documented instances of encroachment on farmland in regions[26]. To rigorously quantify suitable areas for VRE deployment, we adopt a spatially explicit recognition methodology. This approach integrates 300-m-resolution global data on the type of land use with key geographic and technical parameters (including elevation, slope and reserve areas) to determine a pixel-level suitability factor (available-to-total land ratio, 0–1). In this resource assessment, stringent thresholds are applied to restrict cropland conversion, prioritizing the preservation of agricultural land (details in Supplementary Section 2.1). Integrating this detailed spatial assessment into the GISPO model, our analysis indicates a substantial land requirement for solar PV by 2050: over 90,000 km^2 for utility-scale systems. This huge area is comparable to the landmass of a mid-sized nation, for example, Portugal. The integration of distributed solar PV, leveraging an estimated 43,000 km^2 of rooftop surface area, has effectively ameliorated the land-use requirement of PV development. For wind power, while the physical footprint of turbine bases is relatively small (400–450 km^2 globally), the required spacing between turbines results in an occupied area exceeding 1.42 million km^2.

Figure 4 shows the grid-cell-level installation factors for wind and solar, where the installation factor is defined as the ratio of installed capacity to technical potential (0–1). Technical potential represents the installed capacity upper bound at each grid cell as assessed in our resource assessment, accounting for land availability, technical feasibility and technology-specific features (detailed in Supplementary Section 2). The results reveal a global mean installation factor of 0.37 for utility-scale solar PV (25% of cells exceeding 0.98), a higher mean of 0.77 for distributed solar PV (75% of cells exceeding 0.9) and a mean of 0.52 for wind power (onshore and offshore, 25% of cells exceeding 0.99). The model projections highlight acute land competition in densely populated, high-demand regions of East and South Asia. Japan is expected to build up over 80% of its utility-scale solar PV potential and 65% of its distributed solar PV potential, with its western region approaching over 85%. Similarly, China's northern and eastern provinces are projected to install nearly 90% of available rooftop capacity, and India is estimated to exhaust 90% of its rooftop potential. High installation factors of distributed solar PV are also projected for Pakistan (0.95), the Philippines (0.95) and Indonesia (0.89) and several Middle Eastern nations (United Arab Emirates: 0.96; Kuwait: 0.89; Saudi Arabia: 0.83). Detailed power-grid-level installation factors are provided in Supplementary Table 27. Sensitivity analyses confirm that total solar PV area requirements are consistently within the range of 80,000 to 100,000 km^2, emphasizing the substantial spatial footprint of solar infrastructure. The expected land and rooftop area for solar PV across scenarios is shown in Supplementary Fig. 77.

Supporting technologies and demand-side management

Low-carbon firm generators, such as dispatchable hydropower[52] and nuclear power, are critical enablers for integrating large-scale VRE generation[29]. The GISPO model incorporates detailed parameters of existing hydropower facilities and assesses site-specific potential with strict environmental protection considerations[53] for new development using high-resolution geographical and river discharge datasets, therefore optimizing capacity expansion and reservoir dispatch at the dam-site level. The global hydropower potential with existing installations is estimated at approximately 2,811 GW, with notable concentrations in China (~840 GW), Russia (~244 GW), Brazil (~138 GW), the USA (~128 GW), India (~124 GW), Canada (~120 GW) and Indonesia (~113 GW). Supplementary Section 2.2 provides details for hydropower resource assessment. Figure 5a shows the site-level installed capacity of hydropower and grid-level annual generation optimized by the GISPO in the base scenario, where we project 1,643 GW by mid-century (about 375 GW of new installation), generating around 5.7 PWh yr^{-1} of clean electricity. This installed capacity implies a required net annual capacity increase of roughly 15 GW from 2025 to 2050. Our multiple-scenario analysis indicates the stability of hydropower's installations, with capacities ranging from 1,616 GW to 1,692 GW. Across scenarios, China is the dominant player in the hydropower field, with around 610 GW of capacity generating approximately 2.5 PWh yr^{-1}. These installations are primarily located in its Central (240 GW), Xizang (148 GW) and South (134 GW) regions, consistent with previous regional studies[52]. Major contributions are also seen from Brazil (100 GW, 260 TWh yr^{-1}), Russia (99 GW, 410 TWh yr^{-1}), Canada (81 GW, 260 TWh yr^{-1}), the USA (76 GW, 260 TWh yr^{-1}) and India (73 GW, 200 TWh yr^{-1}). We further examine a scenario without hydropower capacity expansion (WithoutHPExpansion), whereby only existing installations are available for operational optimization. Results demonstrate hydropower's critical role in providing system flexibility: although generation mix and SCOE exhibit minimal deviation from the base scenario, total energy storage capacity needs to increase by 10% to 24,624 GWh to compensate for the loss of dispatchable hydropower resources.

Nuclear energy's role is poised for expansion, with a current global installed capacity of over 390 GWe (gigawatt electrical), generating approximately 2.6 PWh of clean electricity per year[54]. Major nuclear power countries are advancing the miniaturization of next-generation reactor designs, such as China's Generation IV system, to enhance their contribution to future net-zero electricity systems. We define regional capacity limits by downscaling the upper-bound projections of future national nuclear power installed capacity, using the installation share forecast provided by the International Atomic Energy Agency (IAEA)[48], to the power-grid level. The base scenario indicates a global nuclear power deployment of around 1,335 GW by 2050, supporting over 10.0 PWh yr^{-1} of electricity generation. This installation needs an

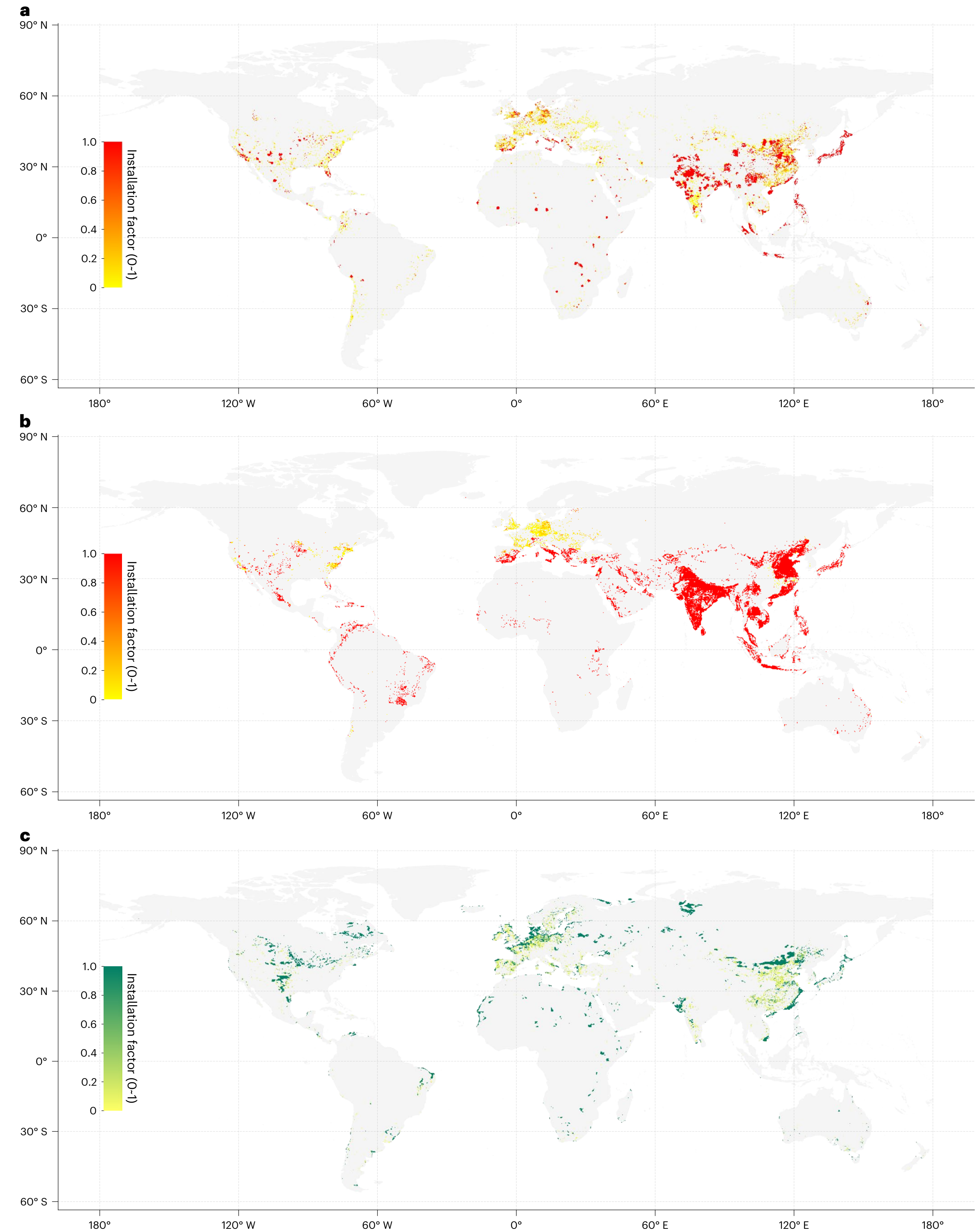

Fig. 4 | Cell-level installation factors for solar photovoltaic and wind power in the base scenario. a–c, utility-scale solar photovoltaic (**a**), distributed solar photovoltaic (**b**) and wind power (**c**). The installation factor (0–1) is defined as the ratio of deployed capacity to the technical potential at each grid cell. Basemap data from Natural Earth (https://www.naturalearthdata.com/).

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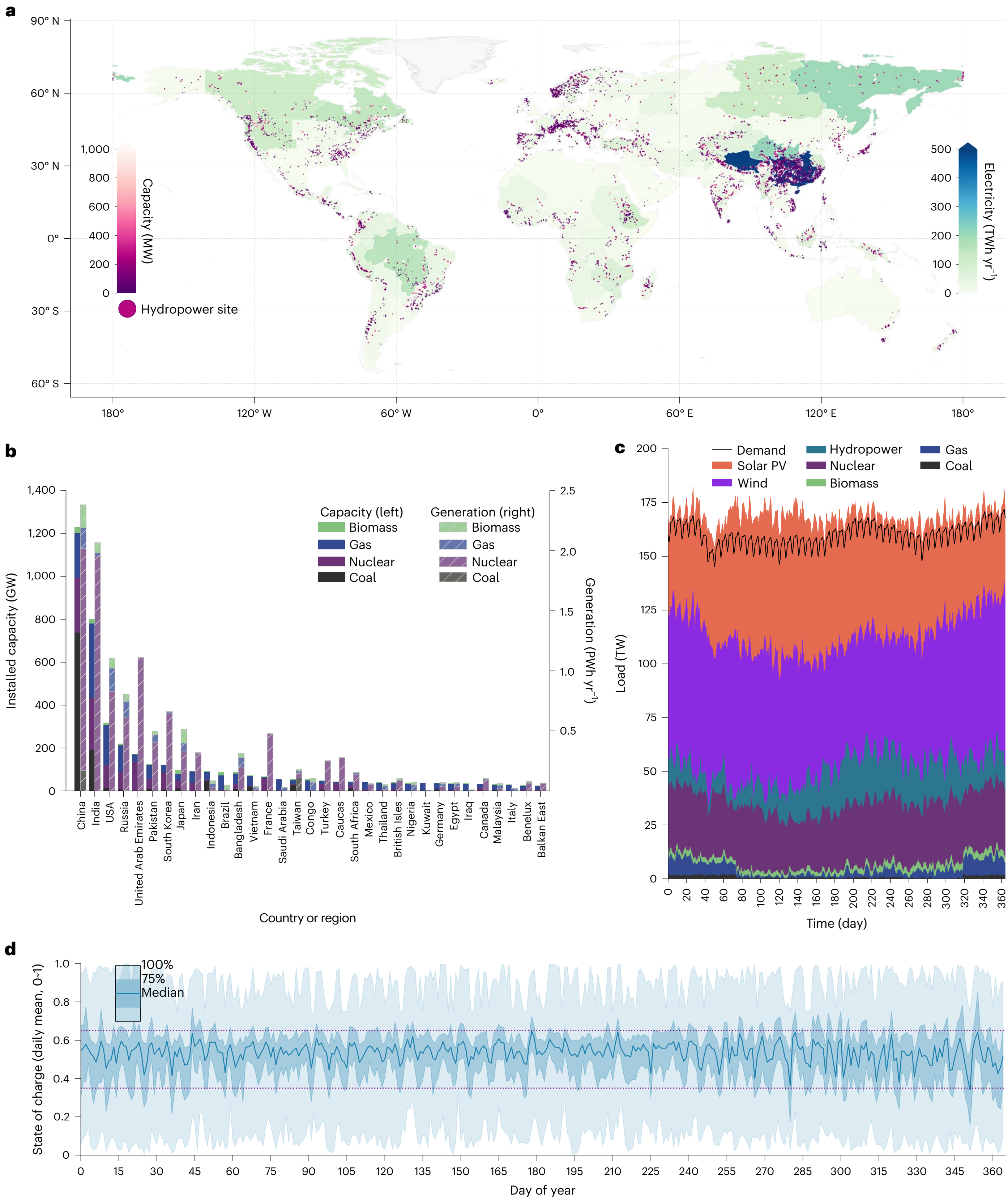

Fig. 5 | Results of installed firm generators and daily generation profile. **a**, Dam-level installed capacity (megawatt) and grid-level generation (terawatt-hour per year) of hydropower. **b**, Capacity (gigawatt) and generation (petawatt-hour per year) of thermal and nuclear power. **c**, Daily power generation and demand profile. **d**, Distribution of daily state of charge for energy storage. Solid lines indicate the median values across all modelled power grids, while the dark and light shaded bands represent the 75th percentile and the full distribution range (100th percentile), respectively. Basemap data in **a** from Natural Earth (https://www.naturalearthdata.com/).

increase of approximately 235% from current levels. Multi-scenario analyses corroborate robust use of nuclear power, showing an installed capacity range of 1,283 GW to 1,338 GW. Future nuclear capacity (including existing installations) is expected to be predominantly located in China (256 GW, 1.85 PWh yr^{-1}), India (243 GW, 1.95 PWh yr^{-1}), the United Arab Emirates (137 GW, 1.1 PWh yr^{-1}) and the USA (105 GW, 0.83 PWh yr^{-1}). We designed a more conservative nuclear expansion scenario (SlowerNuclearExpansion) by capping global capacity below 1,000 GWe with correspondingly reduced regional deployment limits. Relative to the base scenario, this constraint requires additional energy storage (+8.8%) to accommodate expanded VRE generation (solar +7.3%, wind +4.7%), compensating for reduced nuclear output, where these adjustments increase system costs by a minimal amount (<2%). Fig. 5b illustrates the installed capacity and annual generation of nuclear, coal, natural gas and biomass in major countries.

Figure 5c shows the daily electricity supply and demand profiles across global power areas in the GISPO model's planning year, aggregated from hourly dispatch results. The output of firm generators, comprising hydropower, thermal and nuclear sources, is shown beneath the wind power contribution (purple). These firm generators are essential not only for meeting basic electricity and winter heating demands (via CHP) but also are critically important for managing the system variability raised by wind and solar power. Their regulatory capacity allows for output reduction during sudden demand decreases to prioritize the integration of VRE (for example, around days 40 and 80), and conversely, for increased generation to meet demand when VRE output is insufficient (for example, between days 200 and 280) or to compensate for VRE generation dips (for example, solar dip around days 130–180). Furthermore, firm generation flexibility, including adjustments in nuclear or hydropower output (for example, between days 320 and 340) to support the integration of BECCS for emissions targets. Their role extends to providing crucial reserve capacity for grid resilience against extreme events. Energy storage systems (ESS) are indispensable components of future power grids dominated by renewable sources, addressing temporal mismatches between generation and demand and providing essential grid flexibility by providing reserves[31]. Energy storage capabilities within the model are represented by pumped hydro storage (PHS) as a proxy for mechanical storage and batteries (BAT) as an exemplar electrochemical technology. Recognizing the distinct geographical prerequisites for PHS deployment (for example, elevational requirements), we undertake a comprehensive global assessment of its technical potential based on existing geographical information work[55]. Detailed technical and economic parameters for storage systems, along with their assessed potentials, are provided in Supplementary Section 3.6. In the base scenario, global ESS deployment is estimated at 4,943 GW (22,000 GWh), about one-quarter of the total VRE installation (20,000 GW). The projected ESS portfolio is bifurcated into 607 GW of PHS, which is modelled with an eight-hour duration (4,856 GWh) and 4,336 GW of battery storage, assuming a four-hour duration (17,344 GWh). Geographic distribution analysis reveals China as the principal contributor to PHS capacity (354 GW), distantly followed by the USA (45 GW), India (44 GW) and Japan (31 GW). Battery storage deployment exhibits a strong correlation with solar PV integration efforts, concentrated in countries with abundant solar resource potential. India commands the largest share of global battery storage (1,259 GW, representing 30% of the total), deployed in conjunction with 3,127 GW of solar PV. Other leading countries include China (733 GW), Saudi Arabia (317 GW), Indonesia (257 GW), the USA (208 GW) and Pakistan (128 GW). On a global scale, storage systems undergo an average of 286 operational cycles per year (Supplementary Fig. 78 provides energy capacity and annual discharge capacity). An examination of daily state-of-charge (SOC) distributions across grid regions (Fig. 5d) indicates moderate seasonal variability globally; median SOC values are typically between 0.4 and 0.6 annually, with interquartile ranges (25%–75%) extending from 0.35 to 0.65.

Transmission infrastructure, including cross-border transmission, is essential to balance power supply–demand and to manage the spatio-temporal variability and intermittency inherent in weather-dependent renewable energy sources[23,24,28,56–58]. Our GISPO model strategically incorporates these critical components. Existing transmission line data, including location and voltage information, is initially extracted from OpenStreetMap[59], a comprehensive global repository of infrastructure. Thermal capacity for each line is subsequently estimated using methods detailed in refs. 60,61, which accounts for line length and voltage level. Within the model's architecture, a pipeline framework, consistent with established practices in power system modelling, is employed to aggregate transmission capacity between power grids[24,28,56]. On the basis of the network topology, the GISPO model permits expansion of cross-border transmission capacity only between countries with existing interconnections. Domestic transmission expansion is more flexible, allowing for the construction of new corridors both between grids that are already connected and between any geographically contiguous grids, regardless of current capacity. The estimated existing transmission network and the resulting topology are documented in Supplementary Section 3.5. Costs for transmission are modelled as a function of both capacity and line length, with inter-grid distances calculated using the geometric distance between the central points of the load-centre enclosures within each power grid.

There is a critical need for substantial expansion of international transmission networks to support future VRE-dominated grids. A total of 6,300 TW-km of transmission capacity will be needed to achieve a globally accessible net-zero power system by mid-century under the base scenario; of this total, 3,474 TW-km is newly built infrastructure. Cross-border transmission will have to account for 3,200 TW-km (51%) of total transmission capacity, highlighting the importance of international collaboration. Of this needed cross-border capacity, new construction has a share of 73% (2,532 TW-km). The primary locations for these new cross-border lines are projected to be in developing regions, including key links such as Iran–Pakistan (153 TW-km, with 147 TW-km being new), West Africa–Southern Africa (134 TW-km, with 133 TW-km being new) and Central Africa–Congo (116 TW-km, with 115 TW-km being new). The overall capacity factor (annual electricity transfer divided by theoretical maximum capacity transmission over 8,760 hours) of the interconnected global transmission system is projected to be 45%. Our model captures the economic relationship between transmission flows and nodal price spreads (hourly shadow prices from the demand constraints at each grid): when lines operate below capacity, price differentials remain minimal and increase proportionally with flow as larger spreads justify higher marginal transmission costs, whereas congestion causes price spreads to increase substantially as flows reach capacity constraints (the relationship between electricity transmission flow and nodal marginal cost of electricity supply is shown in Supplementary Figs. 142–151). Figure 6a,b provides a spatial representation of net power input (negative values denote export) for each grid region and illustrate inter-grid power flows in the base and LimitedTxExpansion scenario, respectively. Regions exhibiting the highest annual net power imports include East China (1,760 TWh yr^{-1}), Central China (860 TWh yr^{-1}), Northern Pakistan (680 TWh yr^{-1}) and the Mid-Atlantic of the USA (503 TWh yr^{-1}). Conversely, major power-exporting regions are largely situated in China, with North China (1,306 TWh yr^{-1}), Xizang (743 TWh yr^{-1}) and Northwest China (604 TWh yr^{-1}) serving primarily East and Central China. Other major exporters include North Africa (692 TWh yr^{-1}), Russia Ural (652 TWh yr^{-1}), West Africa (644 TWh yr^{-1}) and Texas (623 TWh yr^{-1}).

Globally interconnected transmission networks also facilitate the mitigation of seasonal electricity supply–demand imbalances. Upon disaggregating the net electricity flow maps by season (Supplementary Figs. 81–82), we observe that while most regional electricity exchanges maintain flow directions consistent with their

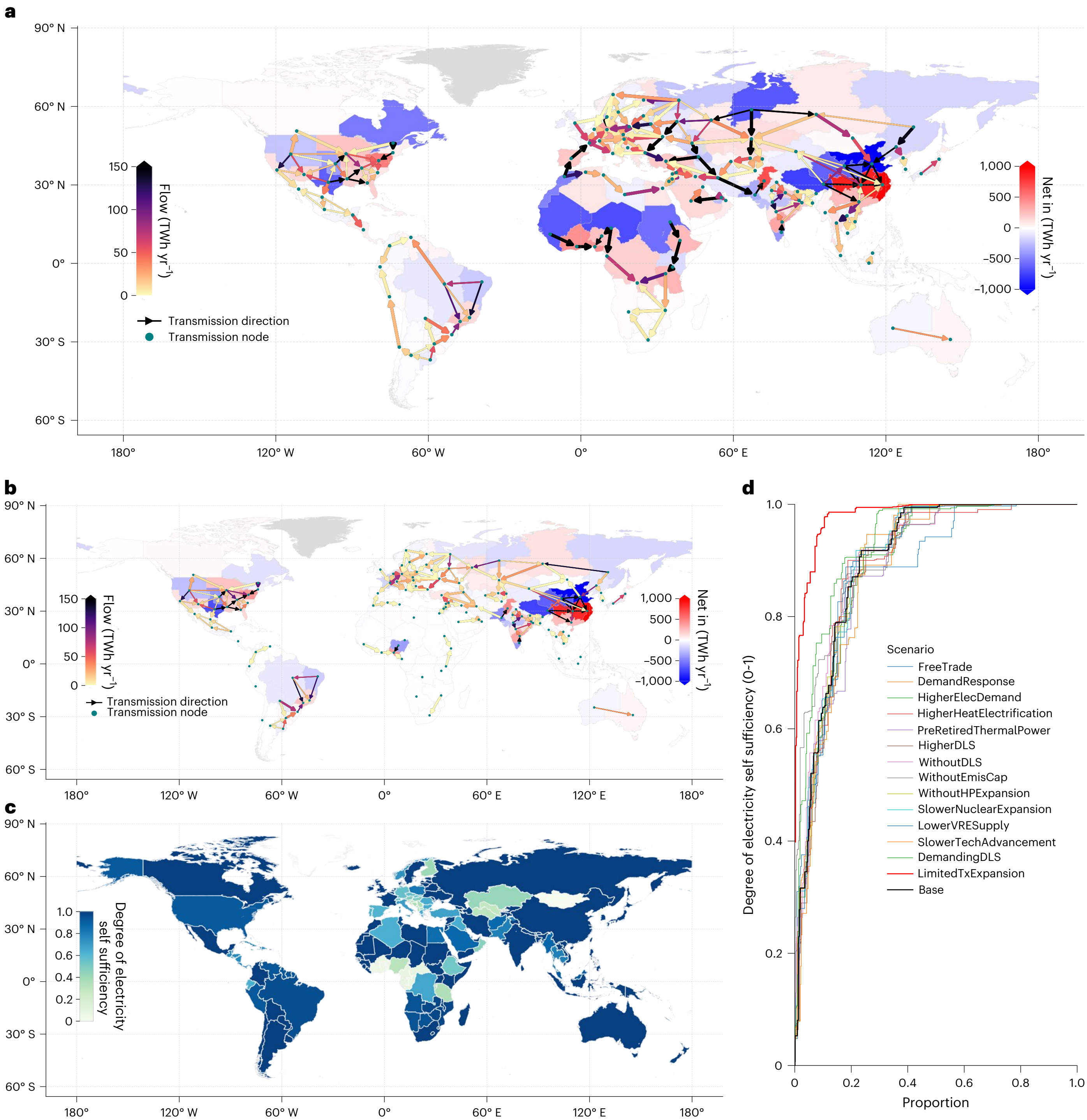

Fig. 6 | Results of long-distance transmission and energy storage. a,**b**, Annual electricity flow (terawatt-hour per year) alongside long-distance transmission (arrow plot) and net transmitted-in electricity for each power grid in the base (**a**) and LimitedTxExpansion scenario (**b**). The negative value means net transmitted out. Wider arrowed lines represent cross-border transmissions. **c**, Degree of electricity self-sufficiency for each power region (country level). The metric here is defined as the ratio of electricity generated by local power to the total demand; the value 1 means local generators can meet the electricity demand entirely. **d**, Cumulative distribution of degree of electricity self-sufficiency in each scenario, weighted by annual local electricity demand. Basemap data in **a**–**c** from Natural Earth (https://www.naturalearthdata.com/).

annual patterns, notable seasonal flow reversals occur. For instance, although Vietnam exports net electricity to South China on an annual basis (Vietnam to South China 196 TWh yr^{-1}; South China to Vietnam 185 TWh yr^{-1}), China's electricity flows into Vietnam through the Vietnam–South transmission line during spring (85 TWh net in) and autumn (15 TWh net in). Similar seasonal bidirectionality is also observed in Central Asia, North America and South America, further underscoring the critical importance of regional electricity interconnection.

The widespread availability of renewable energy sources, in contrast to the uneven distribution of fossil energy, translates to a broad potential for elevating electricity self-sufficiency in nearly all nations. We quantify the degree of regional energy independence using the electricity self-sufficiency metric—the ratio of electricity generated locally to the annual power demand. A value approaching unity signifies minimal reliance on external power. Despite significant international transmission capacity installed as discussed above, we still find a mean self-sufficiency of 0.8 in the base scenario, with more than half of the power regions achieving a self-sufficiency level above 0.95 (Fig. 6c). Robustness analysis across multiple scenarios confirmed the stability of these findings, with mean self-sufficiency ranging from 0.77 to 0.88 and the 50th percentile values between 0.95 and 0.99 (Fig. 6d). This highlights the capacity of a net-zero electricity system based on wind and solar to considerably guarantee countries' energy supply from their resources and thereby bolster their energy security.

To assess the impact of premature retirement of coal and gas-fired units on power sector transition, we design a PreRetiredThermalPower scenario in which only CHP units remain operational until 2050, while all other conventional coal and gas generators retire early by 2050. This scenario reveals substantial changes in the global capacity mix. Without constraints from existing coal and gas capacity, the system exhibits a stronger preference for wind power, with global installations increasing by 15.6%. Transmission infrastructure expands by 17% to enhance inter-regional coordination. Solar PV capacity decreases by 13%, with a corresponding 15.5% reduction in storage capacity. These results indicate that wind–transmission combinations prove more advantageous for system flexibility at large regional scales (for example, Eurasia), while dispatchable generation plays a greater role in facilitating solar integration. However, certain regions deploy additional storage for balancing. In Australia, solar PV capacity increases by 13% to 87 GW, accompanied by a 63% increase in battery storage to 104 GWh. North America demonstrates a 47% storage increase to 1,168 GWh, with wind and solar installations only increasing by 5%, while system costs decline by 1% under the early retirement.

Demand response is proven to serve as an effective mechanism for enhancing power system resilience, efficiency and renewable energy integration in regional power system modelling[62]. In this study, we investigate the impact of demand response at the global scale by incorporating it as hourly demand reductions and increments that balance out monthly. With an annual response capped at 5% of total demand, we find that this enhanced flexibility lowers system costs by 6.5% (US$182 billion), achieving a total response of 2.91 PWh yr^{-1}. Cost reductions are realized by decreasing the reliance on ancillary service capacity and minimizing the operational cycling of thermal and nuclear generators, as illustrated by the daily generation profiles (Supplementary Fig. 79). This flexible operation allows for a 4% increase in solar PV capacity and a 4.6% decrease in wind installations. It concurrently reduces the need for PHS by 28%, battery storage by 16% and long-distance transmission lines by 13%. Furthermore, this flexibility lowers renewable energy curtailment rates for utility-scale solar PV (from 4.8% to 3.9%), distributed solar PV (from 3.9% to 2.9%) and wind (from 3.0% to 1.3%). Whereas the practical implementation of the flexibility levels modelled in this study requires further evaluation, demand response clearly demonstrates significant importance in facilitating a just transition driven by renewable energy development.

Higher electrification of heating in the building sector may accelerate emissions reductions globally[63,64]. While our base scenario generally elevates electricity demand during the heating season, we design an additional scenario to test the impact of seasonal load shifts. This scenario, termed HigherHeatElectrification, maintains the same annual total electricity demand as the base scenario but redistributes loads to reflect higher heating electrification rates during the heating season. The scenario incorporates signals including population, temperature and seasonal heating demand patterns, adjusting the seasonal load profile across grid regions under an elevated heating electrification rate of 50%. The detailed methodology and procedures are presented in Supplementary Section 3.1.3, with comparisons between the adjusted loads and base scenario loads shown in Supplementary Fig. 42.

Optimization through the GISPO model reveals that net-zero emissions and DLS balance in the power sector remain achievable, though with modifications to the generation portfolio. Wind power capacity increases by 9.5%, as its consistent generation profile better accommodates the round-the-clock heating demand during the heating season. Conversely, solar PV capacity decreases by 8.6%, while energy storage capacity reduces by 8.3%. Inter-regional power transmission becomes more critical under this scenario, increasing by 7.0%, and SCOE rises by 4.5%.

Carbon storage and marginal abatement cost of CO_2

In this study, we constrain the net-zero emissions targets at the region (for example, China, the USA and India) or country-group level (the European Union), with the solutions to install carbon-free generators (renewables and nuclear power) and deploy carbon capture and storage (CCS, new plants and retrofit conventional thermal power). The modelling framework for CCS employs spatially resolved CO_2 source–sink matching with capture, transportation and injection costs. A conservative assessment of global bioenergy potential, strictly limited to residues from agricultural, forestry and grassland to avoid potential side effects, yields approximately 200 EJ biofuel potential annually. For DACCS, the model incorporates four distinct sorbent material types, with captured CO_2 directed to either onshore or offshore deep sedimentary aquifers for permanent storage. Our geospatial analysis estimates global CO_2 storage capacity at about 3,600 $GtCO_2$, consistent with previous studies[65]. The transport for CO_2 captured to sequestration sites is optimized through the integration of distance-dependent costs governing pipeline routing. We show details of the resource assessments, technical parameters and cost assumptions in Supplementary Sections 3.7 and 3.8.

Figure 7 illustrates the optimal spatial configuration of CO_2 capture and storage infrastructure, along with regional marginal CO_2 abatement costs and power sector fossil fuel emissions relative to capture rates. We find that the remaining thermal power (coal and gas) results in annual emissions totalling 1,000 $MtCO_2$, of which about 100 $MtCO_2$ is captured by CCS. These thermal power emissions comprise 249 $MtCO_2$ from coal (primarily in China, ~150 Mt) and 751 $MtCO_2$ from natural gas (predominantly in the USA, ~100 Mt), largely attributable to the necessary must-run operation of CHP plants for heating loads. Although retrofitting 30 GW of existing coal and 10 GW of gas capacity with CCS equipment is found within the base scenario, these measures are insufficient for comprehensive system decarbonization. To achieve a globally net-zero electricity system by 2050, the model indicates the need for the complementary deployment of 107 GW of BECCS capacity, delivering approximately 890 $MtCO_2\ yr^{-1}$ in net-negative emissions and 10 $MtCO_2\ yr^{-1}$ of DACCS capacity in a few regions with limited biofuel resources, such as Kuwait (4.1 $MtCO_2\ yr^{-1}$) and Saudi Arabia (2.8 $MtCO_2\ yr^{-1}$). Sensitivity analyses indicate that higher electricity demand will lead to an increase in carbon emissions due to greater use of fossil-based peaking capacity, while lower demand will yield a corresponding decrease.

The marginal abatement cost (MAC) of CO_2, defined as the shadow price associated with emissions constraints in the linear optimization, represents the incremental cost per tonne of CO_2 reduction required to satisfy system objectives (net-zero emissions and global decent living standard), expressed in US$ t^{-1} CO_2. Our analysis indicates that for most countries in the base scenario, MACs range from US$80 to US$120 $t^{-1}CO_2$. This broad consistency in marginal abatement costs across diverse regions suggests that there are common characteristics in the challenges of achieving a net-zero power sector, even where investment costs vary. However, a distinct group of countries ex- hibits substantially higher MACs, many exceeding US$200 t^{-1} CO_2. These include Iceland (US$294 t^{-1} CO_2), Kuwait (US$270 t^{-1} CO_2), Saudi Arabia (US$213 t^{-1} CO_2), United Arab Emirates (US$206 t^{-1} CO_2), Oman

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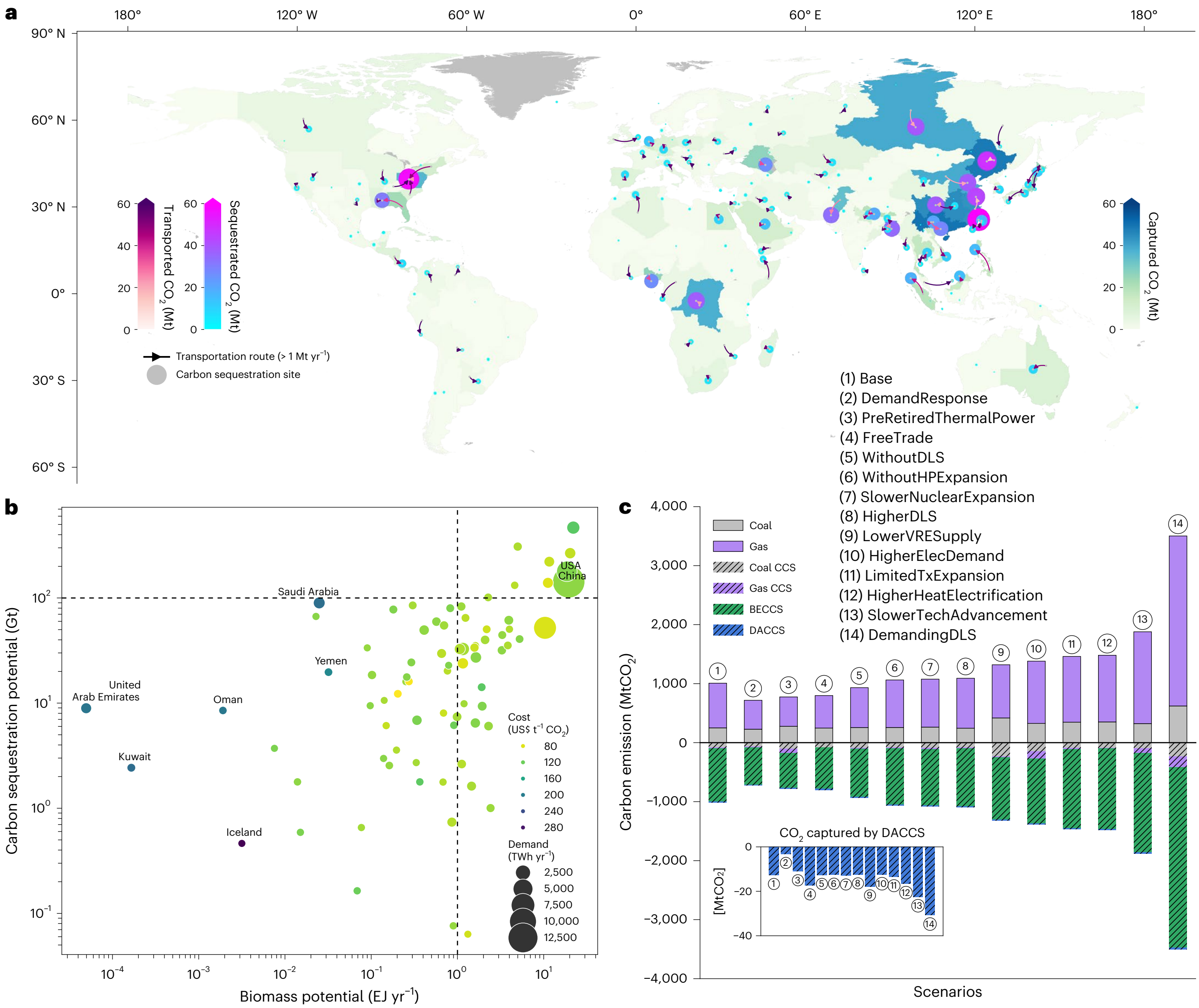

Fig. 7 | Results of carbon capture and storage deployment, carbon abatement costs, and marginal carbon abatement cost. a, Annual CO_2 flows (megatonne per year) in the power sector for the base scenario in 2050. Regional polygons show captured CO_2 from thermal plants or direct air capture (DAC, right). Circles indicate CO_2 injected volumes at storage sites. Arrows depict source–sink transport routes, with thickness (or shade) representing transported volume (left). **b**, Marginal CO_2 abatement cost (MAC) in US dollar per tonne CO_2 by country/region for the power sector in 2050 under the base scenario (coloured according to the associated colour bar). **c**, Annual carbon emissions MtCO_2 from thermal power plants across scenarios. BECCS: bioenergy with carbon capture and storage; CCS: carbon capture and storage; DACCS: direct air capture with carbon storage. Basemap data in **a** from Natural Earth (https://www.naturalearthdata.com/).

(US\$204 t^{-1} CO_2) and Yemen (US\$190 t^{-1} CO_2). As depicted in Fig. 7b, these high-MAC countries share a common set of limitations: (1) scarce biomass fuel potential (less than 0.04 EJ yr^{-1}); (2) restricted CO_2 storage potential, which hampers CCS installation and (3) constrained or non-existent power interconnections to neighbouring systems. Although BECCS deployment is constrained by sustainability considerations, as implemented with conservative biomass availability estimates, it remains the most cost-effective strategy for achieving net-zero emissions. Regions without sufficient biomass resources for BECCS necessitate the deployment of DACCS, which generally incurs substantially higher costs, thereby driving up MACs. Whereas carbon emissions targets are typically enforced at national or regional levels, countries facing abatement challenges can potentially mitigate these pressures by importing low-carbon electricity from regions possessing abundant mitigation resources.

Investments and returns

Effective investment and the anticipation of reasonable financial returns will be key to achieving a successful energy transition and promoting electricity equity. We quantify the requisite investment for power generators using the annual capital expenditure (assuming a 7.4% for weighted average cost of capital) and operational cost and estimate the expected returns by the products of the electricity generation and marginal cost (MC; US dollar per megawatt-hour) of electricity supply. Within the linear optimization framework of the power system model, the hourly marginal cost of electricity supply is derived from the dual values (shadow price) associated with the power balance constraint[23,66–68]. In a competitive power market, the clearing price is theoretically equivalent to the MC of electricity supply under efficient operation[68]. Figure 8a illustrates the hourly distribution of MC across each modelled power grid over the 8,760 hours

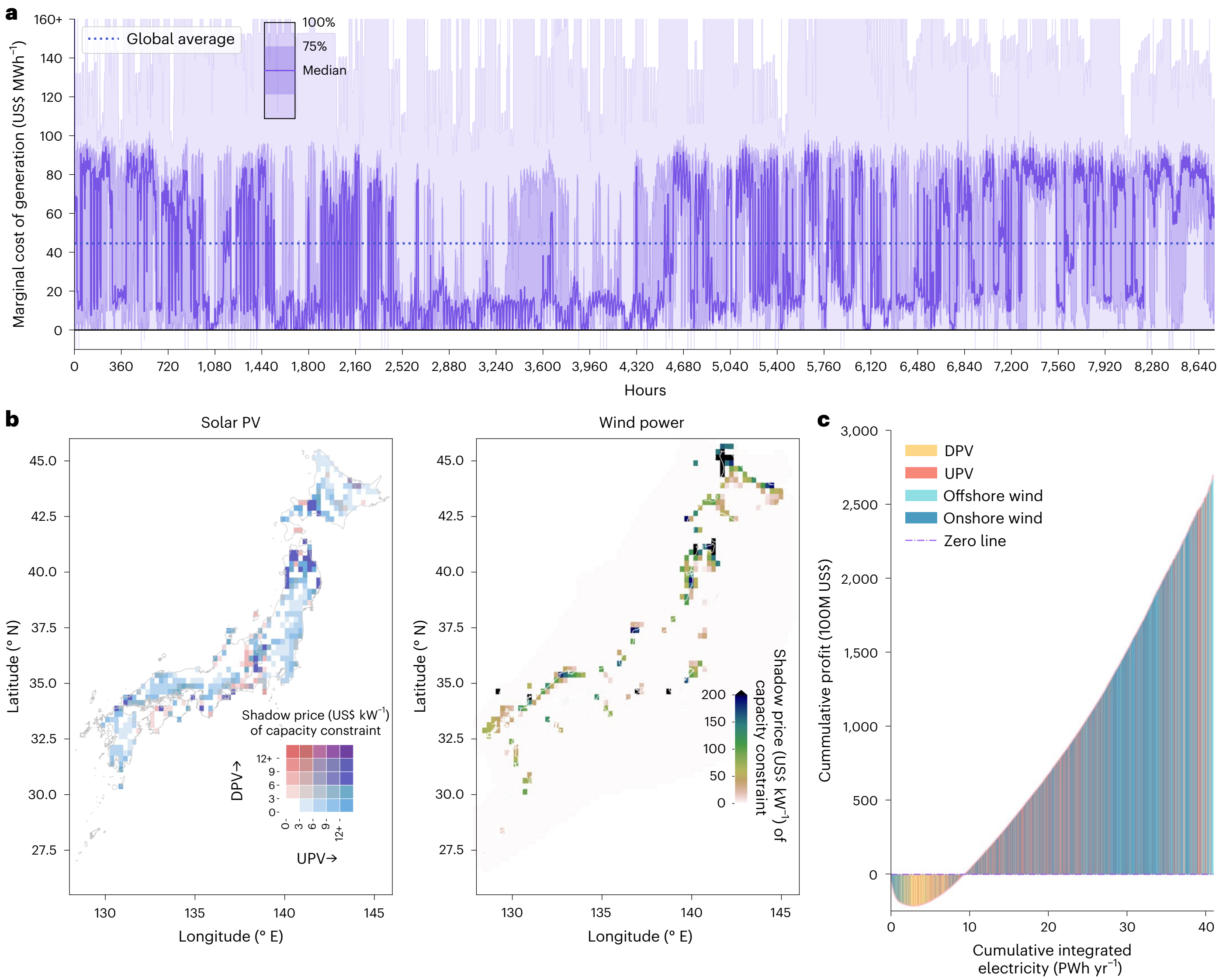

Fig. 8 | Economic performance and revenue sufficiency in the mid-century power system. a, Hourly marginal cost of generation (US dollar per megawatt-hour) distribution of power grids through the full 8,760 hours in the planning year. Solid lines indicate the median values across all modelled power grids, whereas the dark and light shaded bands represent the 75th percentile and the full distribution range (100th percentile), respectively. **b**, Spatial distribution of the annual shadow price (US dollar per kilowatt) associated with wind and solar deployable capacity constraints in Japan. High shadow prices indicate locations where resource availability limits further expansion, thereby generating inframarginal rents. **c**, Relationship between cumulative integrated electricity supply (petawatt-hour per year) and annual economic profit for wind and solar photovoltaic technologies globally (DPV: distributed photovoltaic; UPV: utility-scale photovoltaic). Profit is defined as total market-based revenue (marginal cost of electricity supply multiplied by post-curtailment generation) minus the total annualized cost. The horizontal segments represent the volume of generation occurring at different profitability levels. Basemap data in **b** from Natural Earth (https://www.naturalearthdata.com/).

in the base scenario (continent-level distribution in Supplementary Fig. 83). Given the near-zero marginal cost of VRE sources, the integration of large-scale renewables will lead to substantial fluctuations in system MCs. The frequent occurrence of both exceptionally high MCs (>US$160 MWh^{-1}) and near-zero MCs (~US$0) highlights important system volatility projected for the mid-century. Intriguingly, negative MCs of generation are also observed during specific time steps, as shown in Fig. 8a. We show the check on the relationship between MCs and investment costs, VRE integration shares, flexible demands and electricity transmission in Supplementary Figs. 115–151.

We assess revenue sufficiency as the difference between annual system expenditure and market-based revenue, evaluating whether the system operates at a surplus or a deficit under marginal-cost pricing. On the basis of this metric, our analysis reveals that aggregate system revenue is insufficient to cover total expenditure (including annual capital expenditure for existing generators if they are still in use) in the base scenario, resulting in an annual deficit of approximately US$135 billion and a net profit margin of −4.7% yr^{-1}. This missing money problem underscores that marginal-cost pricing alone may fail to ensure full cost recovery in low-carbon systems. An examination of technology-specific economics indicates a wide dispersion of annualized returns on investment. Negative returns are predominantly concentrated among technologies providing system essential reliability services, stability and emissions compliance, including battery storage (−0.1%), pumped hydro storage (−0.1%), natural gas power (−1.5%) and coal power (−2.4%). Losses for fossil fuel generators primarily reflect economic premature retirement or stranded assets—where sunk capital costs cannot be recovered through dwindling operating hours under stringent carbon constraints.

In contrast, VRE and nuclear power exhibit favourable returns of 9.4% and 8.0% under the strict net-zero emissions constraint, respectively. VRE profitability arises primarily from binding resource availability constraints, which generate positive shadow prices in our high-resolution grid-cell representation. At locations where renewable capacity reaches the deployment upper bound, positive shadow prices create inframarginal rents that reflect the system value of limited access to premium wind and solar resources. This mechanism—absent in aggregated regional models where capacity constraints rarely bind—highlights that resource availability (for example, land use), rather than solely technology cost, becomes a defining economic driver in future renewable-dominated systems. To validate this, we use Japan as a case study to derive the cell-level shadow price of the VRE capacity constraints and design a counterfactual experiment where we relax the capacity bound (CapBoundRelax). In the base scenario, we find that the shadow prices of the newly built cells with positive profits range from tens to hundreds of US dollars per kW, as shown in Fig. 8b. When capacity-bound constraints are relaxed, VRE capacity concentrates in a few high-quality wind cells, eroding VRE profitability (Supplementary Fig. 60). Nuclear power achieves positive returns from binding capacity constraints based on IAEA deployment projections, indicating that nuclear's role in decarbonization may be underestimated when such realistic constraints are not considered. The positive profits reflect its carbon-free profile and low operational costs to maintain high capacity factors within renewable-heavy generation mixes. These divergent economic outcomes underscore critical imperatives for electricity market design, particularly the need for effective mechanisms (for example, capacity market[69]) that ensure adequate cost recovery and incentivize necessary investment across the diverse portfolio of assets.

To quantify the economic performance of VRE at a more granular level, we calculate annual revenues for each grid cell by multiplying post-curtailment generation by the prevailing nodal electricity price. Net profits are subsequently derived by subtracting annualized capital costs and operational expenditures. As depicted in Fig. 8c, the cumulative integrated electricity supply and corresponding financial returns vary across VRE types in the base scenario. Our projections indicate that global VRE systems could generate an estimated US$271 billion in cumulative profits under our cost-optimized deployment trajectory. Approximately 6% of total generation (2.5 PWh yr^{-1}) occurs at nodal prices near or below zero, representing marginal losses. This overall economic viability is sustained by the substantial profitable generation volume (7.5 PWh yr^{-1}), which offsets these losses. Although the life-cycle profitability of VREs is generally acceptable, the inter-annual revenue volatility of VRE assets could hinder VRE investment in the real world. Long-term contracts could be a key mechanism for managing financial stability and potentially redirecting revenues towards grid-balancing technologies[70].

International collaboration

We further explore two important international collaboration options for achieving net-zero power systems: international grid infrastructure connection and free trade. We demonstrate that benefits would be substantial, especially for countries in the global south. For international grid connection, a sensitivity analysis termed LimitedTxExpansion examines the impacts of no-new expansion for cross-border transmission lines on power systems to avoid potential geopolitical risk and system security. In this scenario, we find that limited cross-border transmission expansion will result in an approximately 43% decrease in total inter-grid capacity to 3,600 TW-km, and increase system cost by 5.6% (about US$157 billion). These capacity reductions are most pronounced in Africa, leading to aggravated challenges for energy equity and a 20% increase in total electricity costs, comparable to previous regional modelling results[71]. This scenario also demonstrates the relationship between storage and transmission infrastructure, both complementary and substitutable dynamics, where the battery storage capacity expands by 18% (+770 GW), primarily driven by deployments in Africa (+497 GW). Notably, the LimitedTxExpansion scenario results in higher regional self-sufficiency (25th/50th percentile 0.87/0.99) compared to the 0.6–0.7 range observed in other scenarios (Fig. 6c,d).

Existing international trade barriers, particularly those impacting renewable energy technologies and batteries, pose substantial impediments to technological diffusion[72,73]. Conversely, this analysis reveals the important potential of trade liberalization in facilitating a globally equitable transition to a net-zero electricity system[74]. Employing a comparative scenario modelling approach, we project 2050 investment costs for renewable technologies under two distinct frameworks detailed in Supplementary Section 3: a base scenario incorporating country-specific renewable investment costs informed by uniform global learning curves and a FreeTrade scenario postulating the adoption of the anticipated lowest investment costs (projected for China) universally across all regions. Our findings demonstrate substantial economic benefits stemming from trade liberalization, with average global electricity generation costs projected to decrease by 12.2% under the FreeTrade scenario, from US$49 MWh^{-1} to US$43 MWh^{-1}. This projected cost reduction corresponds to approximate annual global savings of US$345 billion by 2050, representing approximately 26.5% of current annual worldwide power sector investment levels[75]. Regional analyses further reveal disproportionate benefits, with cost reductions exceeding 10% projected for both developed economies and various African regions. Specifically, compared to the base scenario, most African nations exhibit projected annualized cost declines exceeding 20%, including Egypt (−22.5%), Nigeria (−24%), Central Africa (−25%) and West Africa (−37%). Substantial cost reductions are also evident in developed markets, such as Japan (−31%), Finland (−25%), Germany (−24%), the USA (−16%) and Canada (−14%).

Discussion

In this study, we present a temporally and spatially resolved power system model that co-optimizes capacity expansion and hourly dispatch over a full year (8,760 hours) to delineate the pattern of global net-zero power systems while explicitly addressing electricity equity. Our analysis identifies system architectures capable of meeting a minimum electricity standard of 2,500 kWh yr^{-1} per capita globally. Achieving this target necessitates a massive deployment of renewable energy and associated infrastructure, including approximately 20,000 GW of wind and solar PV capacity, 1,650 GW of hydropower, 1,300 GW of nuclear power, 22,000 GWh of energy storage and 6,300 TW-km of inter-grid transmission capacity. Critically, leveraging advancements in renewable energy technology and supporting infrastructure, our findings demonstrate that it is not only technically feasible but also economically attractive to achieve net-zero emissions globally, while simultaneously ensuring access to electricity for decent living standards. Across all modelled scenarios, the simulated average global electricity costs are lower than the level in 2022, underscoring the potential for cost reductions inherent in the transition. These results carry profound implications for global energy planning, spanning land-use policies, trade policies, grid-integration strategies and investment mechanisms required to realize universally accessible, net-zero power systems.

One of the most critical bottlenecks identified in the transition is the increasing tension over land use, particularly in densely populated regions. Our spatially explicit analysis reveals that realizing the required solar PV deployment by 2050 will necessitate over 90,000 km^2 of land—an area comparable to that of a mid-sized nation such as Portugal. This substantial spatial footprint highlights the urgent need for innovative and multi-functional land-use planning that effectively balances renewable energy deployment with competing demands for agricultural production and biodiversity conservation. Our analysis further reveals that substantial scarcity rents could emerge at capacity-constrained high-quality sites. Therefore, regions facing acute land scarcity, exemplified by our findings showing near exhaustion of conventional rooftop solar potential in nations such as Japan and India,

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must proactively explore alternative siting strategies. Future research can assess the utilization of brownfields, vertical spaces, agrivoltaics or floating PV systems to mitigate conflicts.

Grid integration and the expansion of long-distance transmission infrastructure represent another cornerstone of a sustainable power system transition, yet our analysis reveals a notable tension between the geographic distribution of renewable resources and load centres. Counterintuitively, our findings indicate that over 80% of the optimally sited wind and solar PV capacity is located within 200 km of existing load centres. This trend challenges the conventional wisdom that emphasizes exploiting remote, resource-rich areas, such as the Sahara Desert, despite its vast potential, which only contributes about 3% to the global optimal solar PV capacity in our model. Long-distance transmission remains indispensable for balancing regional supply–demand disparities and enhancing system resilience; its efficiency and availability must be improved through advanced grid technologies and reinforced international cooperation. The increase in annual system costs (+5.6%, approximately US$157 billion per year) observed in the scenario that limits new cross-border transmission capacity further underscores the critical role of robust cross-border in- terconnection in achieving cost-effective and sustainable development outcomes. These insights highlight the importance of tailoring grid expansion strategies to regional resource and demand profiles while simultaneously strengthening international cooperation to enhance overall system cost-effectiveness and reliability.

Our analysis of investment dynamics reveals starkly contrasting economic profiles across different technology classes, necessitating innovative market designs to ensure financial viability and equitable deployment. VRE systems exhibit relatively robust estimated returns of approximately 9%, largely driven by declining capital expenditures and near-zero marginal generation costs. In contrast, essential regulatory and infrastructure technologies, such as energy storage and transmission, mostly face financial losses, with estimated negative returns exceeding −9%. This divergence, which reflects the difference between the high market value captured by VRE and the system-balancing costs (or integration costs) borne by flexibility providers[40,76], presents a fundamental challenge for market design: how to reconcile the disparate economic realities of energy generators, flexibility providers and critical grid infrastructure components. Our findings suggest that mechanisms (such as long-term contracts) need to be introduced to redistribute revenues from profitable VRE generation to critical, loss-making grid-balancing technologies, thereby stabilizing overall system economics and incentivizing necessary investments. Furthermore, our analysis highlights potentially superior investment returns in developing economies, suggesting that strategically coupling capacity expansion in underserved regions with robust revenue-sharing and risk mitigation mechanisms could simultaneously accelerate energy access and attract crucial private capital. These findings collectively call for a paradigm shift in energy finance, advocating for integrated approaches that explicitly align profitability, equity and long-term system sustainability.

Our modelling demonstrates that liberalizing international trade in renewable energy technologies could serve as a potent mechanism for accelerating equitable global decarbonization. Free international trade explored in our scenarios yields global average electricity cost reductions of approximately 12.2%, translating to annual savings of around US$345 billion compared to more restricted trade environments. These economic benefits are particularly pronounced in developing economies and across the African continent, where some nations could experience cost reductions exceeding 20%. Such outcomes suggest that dismantling trade barriers not only facilitates the rapid diffusion of clean energy technologies but also enhances energy justice by improving the affordability of clean energy in underserved regions. Realizing these substantial benefits, however, is contingent upon overcoming geopolitical resistance, balancing impacts on local industry development and employment opportunities and fostering collaborative international frameworks that prioritize collective sustainability goals over narrow nationalistic interests. By quantifying the transformative potential of trade liberalization, this study provides a strong impetus for renewed international dialogue aimed at aligning renewable energy deployment policies with broader climate-equity objectives.

Achieving a global power system simulation with full chronological hourly resolution (8,760 hours) represents a substantial methodological advancement, pushing the boundaries of current modelling capabilities. This study is designed to undertake a detailed temporal and spatial analysis of power system transitions on a worldwide scale. The immense computational and data requirements of building and running a model at this resolution inherently lead to certain necessary simplifications and boundary conditions. Consequently, this study is subject to limitations, although we believe they do not invalidate the main findings at the aggregate global level. For example, specific technologies, such as geothermal and concentrating solar power, and long-duration energy storage (larger than 10 hours), are not yet modelled. Power-to-X (P2X, for example, green hydrogen and synfuel industry) is not explicitly modelled in this study, leaving it as a direction for future work. Given the lack of location-specific information on rooftop solar PV at the global scale, we do not incorporate the installed capacity as input, which may underestimate some regions' existing capacity. Additionally, global-scale, more diverse load-side forecasting coupled with industrial transformation, and load shifting and demand management, could be further investigated in the future, potentially in conjunction with the GISPO model. A promising future application would involve using computationally efficient models to conduct large-scale systematic uncertainty analysis (for example, thousands of scenarios), with these results then serving as boundary conditions for high-resolution simulations using the GISPO model. Another potential direction is to integrate the dynamics and potential constraints of the global supply chains of key materials. Furthermore, future research should incorporate the implications of cross-border transmission expansion subject to more diverse constraints, such as environmental restrictions and geopolitical risks. To manage the complexity of the global hourly simulation, nonlinear power system operations such as unit commitment and optimal power flow are represented using simplified approaches that have been validated with negligible loss of fidelity[15,23,24,28,57]. The scope is also limited to the power system itself, omitting detailed analyses of broader environmental and socio-economic impacts.

While the influences of climate change on power systems with high penetrations of renewables are a topic of critical importance, this study does not incorporate climate projection modelling for two principal reasons. First, the meteorological driving data from future-oriented Earth System Models often lack the requisite spatio-temporal resolution for our analysis. For instance, their typical spatial resolutions of 0.5° × 0.5° or coarser and temporal resolutions of 3 to 6 hours are insufficient for the ultra-high precision demands of the GISPO model. Although downscaling techniques using machine learning on historical data exist[77], the inherent uncertainties of fitting algorithms led us to prioritize the direct use of authentic historical data. We therefore employed the European Centre for Medium-Range Weather Forecasts Reanalysis v5 (ERA5) dataset, a widely validated historical meteorology dataset that is even used to train high-accuracy weather forecasting models[78]. Second, Earth System Models in the Coupled Model Intercomparison Project Phase 6 (CMIP6) frequently do not report key meteorological variables essential for robust renewable energy assessments. A critical example is wind speed at a 100-metre hub height. Many past studies have resorted to extrapolating this variable from 10-metre wind speeds using the wind power law[22,43], a method that has been shown to introduce substantial bias[79]. Fully addressing these aspects alongside the high-resolution presents substantial modelling challenges but also heralds promising future research.

In summary, the Global Integrated Sustainable Power-system Optimization (GISPO) model, as presented in this work, offers an ultra-high-resolution framework for navigating the complexities of global energy transitions towards a net-zero future. Its spatio-temporal granularity and integrated co-optimization of capacity expansion and operational strategies provide a robust tool for exploring technically feasible and economically viable solutions to universal electricity access under stringent climate targets. As the current study focuses on a representative mid-century milestone year (2050) of the transition of global power systems, we only run the model as a cross-sectional analysis for 2022 and 2050 (focusing on the results in 2050). Future studies can explore the model's potential as a dynamic tool for energy system planning and policy design at the global scale in broader timescales, exploring issues such as land-use tensions, the impact of trade liberalization or restriction, grid connection challenges, the impact of advanced technologies, the resilience of power systems to climate change and strategies to achieve energy justice at global and regional scales. Detailed model outputs can inform integrated spatial planning, guide strategic infrastructure investments and foster international cooperation, providing critical insights for researchers and policymakers aiming to accelerate a just and sustainable energy future.

Methods

Energy resource assessment

Our assessment quantifies global wind (onshore and offshore) and solar (utility-scale PV and distributed PV) resources through the estimation of hourly capacity factors (CF, 0–1) and maximum installation capacity potential (MW). The foundation for this analysis rests on high-resolution meteorological data, which is acquired from the European Centre for Medium-Range Weather Forecasts Reanalysis Version 5 (ERA5) dataset[80]. ERA5 provides comprehensive global climate and weather records spanning over eight decades with a spatial resolution of 0.25° × 0.25° and hourly temporal resolution, including critical variables such as wind speed (at 10 m and 100 m), surface shortwave radiation and ambient temperature. Leveraging these data, hourly CFs for both wind and solar resources are computed for each grid cell using widely accepted generation models[23,28,79,81–83], incorporating wind turbine model parameters from the National Renewable Energy Laboratory[84] and fixed-tilt solar PV generation model from ref. 15. The estimation of maximum installation capacity potential commences with the delineation of suitable deployment areas. For wind power (onshore and offshore) and utility-scale solar PV, suitable areas are defined by applying constraints related to natural/biodiversity reserves, technical feasibility (for example, elevation, slope, water depth and shipping lanes) and land-use categories[23,24,83]. The potential for distributed solar PV is based on the available rooftop area, which necessitates an estimate of global rooftop area in each grid cell. We first compile vectorized rooftop data from all available public sources, which are recognized by satellite imagery and computer vision algorithms[85]. We then predict rooftop areas lacking direct vectorized data with an XGBoost regression model developed and trained using cells with reliable rooftop data, employing population, road length, night light and built-up area as input variables[49,50]. The suitable area for distributed solar PV deployment in each cell is subsequently determined by multiplying a specified fraction (0.35) to avoid overestimating available deployment area and generation potential, adopting the assumption in[50]. The installation potential capacity (MW) within the identified suitable areas (km^2) for wind and solar is then derived by multiplying the suitable area by assumed installation densities ($MW\ km^{-1}$), incorporating assumptions such as a 7 × 7 rotor diameter spacing for wind power installations[86,87] and a fixed-tilt model for solar PV[15]. Details on the methodology, datasets and results are provided in Supplementary Section 2.1.1–2.1.2 for capacity factor assessment, 2.1.3–2.1.4 for the potential area assessment and 2.1.5 for the installation capacity potential assessment.

Beyond wind and solar, this study evaluates additional energy-related resources, encompassing hydropower installation and output potential, available biofuel resources and carbon storage capacity. The hydropower installation potential is established via a least levelized cost of energy (LCOE) analysis that incorporates constraints including inundated area, environmental conservation requirements, population displacement considerations and grid-integration distance, drawing upon methodologies from prior research[53,88]. The LCOE minimization process determines optimal dam height (m), reservoir capacity (m^3) and resultant installation capacity potential (MW), which parameterize subsequent hydropower deployment modelling. Hydropower generation simulations for run-of-river and reservoir types employ 2019 discharge data ($m^3\ s^{-1}$) at a 3-hour resolution[89], assuming constant hourly discharge within each 3-hour interval. For all potential and existing plants, our model mandates that the available water for generation is the natural river discharge after subtracting the environmental flow, which is defined as the 10th percentile of historical discharge (1980–2019) (detailed in Supplementary Information Section 2.2). Biofuel resource potential is quantified by the residues from agricultural, forest and grass to avoid potential side effects in $TJ\ yr^{-1}$ for each pixel (1 km × 1 km and 10 km × 10 km). In the GISPO, biofuel is combusted for biomass energy and BECCS, which are modelled at the power-grid level with a relaxed unit commitment algorithm. Therefore, we aggregate the pixel's resource potential to the power-grid level as the upper bound for the annual available fuel, and adopt a higher fuel cost (US$100 per ton) than coal fuel in resource-abundant areas (for example, about US$50 per ton in China) to reflect the collection and transportation costs for biofuel. Additionally, the installation capacity potential of biofuel power is also constrained according to biofuel potential to avoid overestimating the role of biomass energy, applying assumptions of a 0.35 thermal efficiency[90] and 6,132 annual equivalent operating hours[57]. This yields an estimated biomass installation potential of approximately 4,050 GW; biofuel resource distribution details are provided in Supplementary Section 2.3. Regarding carbon storage, the assessment adopts deep saline aquifers and CO_2-enhanced oil recovery reservoirs[65]. The delineation of sedimentary basin surface areas relies on the World Geologic Provinces dataset, refined using findings from ref. 65. This approach estimates a global CO_2 storage potential of over 3,600 Gt (estimation formula and results are presented in Supplementary Section 2.4).

The Global Integrated Sustainable Power-system Optimization model

The GISPO model employs a hierarchical spatial disaggregation reflecting the organization of global power systems. At the highest level, the world is divided into 91 power planning regions, generally corresponding to individual countries (for example, China, the USA, Russia, India) but occasionally representing multi-country groupings (for example, the Iberian region combines Spain and Portugal).

These regions are further subdivided into 144 power grids globally (for example, East China grid), as specifically mapped out in Supplementary Fig. 2. Each power grid constitutes the elemental unit within the model where the simulation of power supply–demand balancing, generation dispatch and transmission line interactions takes place.

Optimization within the GISPO model targets the minimization of total annualized system cost, encompassing both capital investments and operational expenses, across a full year (8,760 hours) of hourly operation. This optimization process, conducted for each planning year and driven by 2019 meteorological data, is subject to stringent requirements for meeting hourly electricity demand and a diverse set of engineering, economic and policy-related constraints. The model determines the optimal deployment and hourly operational profiles for a comprehensive suite of energy technologies: onshore and offshore wind, utility-scale PV and distributed PV, run-of-river and reservoir hydropower, other firm generators (coal, gas, nuclear, bioenergy), energy storage systems (battery and pumped hydro storage, carbon capture and storage, direct air capture and intra- and inter-grid transmission infrastructure. GISPO provides outputs for each planning year detailing the optimized capacity expansion and the resulting hourly operational state of these technologies. Spatial resolution for technology installation varies: VRE is resolved at the 0.25° × 0.25° grid-cell level, hydropower at the dam-site level and other generators at the power-grid level. The detailed hourly operation simulation considers factors such as available VRE output, hydropower reservoir levels and discharge rates, thermal and nuclear unit commitment statuses, state of charge for energy storage, required spinning reserves, system inertia requirements and power flow across the transmission network, including ultra-high voltage lines.

Grid integration for renewable energies, encompassing both spur and trunk lines, is modelled by associating integration distance and capacity with deployment decision variables, following methodologies established in previous work[23,24,28]. Substation locations with voltage levels exceeding 220 kV are sourced from OpenStreetMap (OSM)[59] and serve as potential termination points for spur lines. Major load centres are identified as urban areas based on 1:50 M zooming data from Natural Earth Organization[91]; an additional load centre from 1:10 M zooming level is included in remote areas (for example, Xizang) to ensure at least one major node is designated in those regions. A minimize-integration-distance algorithm determines the connections among renewable energy sites, substations and load centres (Supplementary Section 3.5.1 provides detailed results). The model endogenously optimizes the matching of carbon capture facilities with storage sites, allowing CO_2 transportation from each power grid to any eligible site. Transportation distance is estimated using the geographical distance between power-grid centres and storage locations (unit costs are detailed in Supplementary Section 3.7). A relaxed unit commitment algorithm governs the dispatch of thermal and nuclear power plants within GISPO. This approach is adopted because previous studies demonstrate significantly higher solve speed and negligible loss (<1%) compared to full unit commitment[56,92]. Similarly, inter-grid transmission operation is optimized using a transportation (or pipeline) model, an approach validated[93] and widely applied in numerous other PSEMs[24,28,56,57,94,95]. Detailed model formulations are presented in the Supplementary Section 4.

Input data and key assumptions

The study framework sets 2022 as the base year and conducts an optimization of the global power systems layout targeting 2050 with a net-zero emissions goal in the electricity sector worldwide. Base year installed capacities for coal, gas, nuclear, wind and solar power originate from the IAEA and the Global Energy Monitor database[96], which furnishes unit-level data on installed capacity, operational status (operational or under construction), remaining lifetime and locations. By 2022, the operational capacity inventory comprises 2,016 GW for coal power (including combined heat and power plants), 2,268 GW for gas power, 366 GW for nuclear power, 81 GW for biomass power, 1,009 GW for onshore wind, 108 GW for offshore wind and 1,022 GW for solar PV. Complementing this, approximately 1,081 GW of installed hydropower (run-of-river and reservoir) at the dam-site level are included from ref. 96, alongside 167 GW of installed pumped hydro storage capacity from ref. 96. Projects under construction for coal, gas, nuclear and PHS are presumed to enter operation before 2050. Within the planning year, installed coal and gas power capacities undergo retirement upon reaching their designed service life or are subject to retrofitting with carbon capture equipment as necessitated by the optimization. Installed capacities for wind, utility-scale solar PV, hydropower (including PHS) and biomass power are assumed to be renewed at the end of their operational lifetimes. Existing distributed rooftop PV systems are projected to reach the end of their design lifetimes before 2050; these are assumed to be fully retired. Further specifics regarding generator installations and the spatial allocation of VRE capacity to grid cells are found in Supplementary Section 3.

In this study, electricity demand and investment costs serve as key exogenous variables. Regional annual electricity demand (typically a country, for example, China, the USA or a union of several countries, for example, West Africa) is established through a two-pronged approach: leveraging existing forecasts based on historical trends and incorporating an energy equity perspective to ensure decent living standards are met. We synthesize future demand projections from various reports and studies, employing linear extrapolation to 2050 where forecasts did not extend that far (Supplementary Section 3.1.1 provides sources). Simultaneously, we calculate a needs-based demand by multiplying UN 2050 population projections by an assumed 2,500 kWh yr^{-1} per capita, a figure grounded in the 15 GJ yr^{-1} energy requirement for less developed regions by mid-century[46] and a 60% electrification rate. The base scenario for each planning region adopts the greater of these two demand estimates, resulting in a global base demand of about 58.3 PWh yr^{-1}, which aligns well with projections from leading IAMs in the ADVANCE database. Hourly load profiles are sourced from ref. 23 for China and the Neo Carbon project[45] for other nations. For investment costs, country or continent-specific data from reports and studies are adjusted using NREL's reported cost reduction curves to project technology costs in each country for the base scenario (further details in Supplementary Section 3).

Model implementation

The GISPO model is a comprehensive and temporally and spatially resolved global power system planning model. It features an hourly supply–demand balance, high spatial resolution of 0.25° × 0.25° for VRE expansion and dam-level detail for hydropower, which enhances the ability to model land constraints, capture the spatio-temporal heterogeneity of resources (for example, reservoir regulation in hydropower) and explore the long-term roles of storage and transmission in renewable integration. The model's extensive scale, approximately 2.4×10^8 rows, 1.4×10^8 columns and 3.5×10^9 non-zero elements, makes direct optimization computationally prohibitive, even with advanced solvers such as Gurobi. We therefore introduce a model-cutting and distributed parallel solution algorithm based on transmission line topology. This algorithm segments the large model into interconnected sub-regional models according to transmission links, allowing for parallel processing to improve solution efficiency. The method ensures that regions without information exchange (for example, North America and New Zealand) are separated, whereas potentially interconnected areas (for example, Eurasia and North Africa), where power transmission influences outcomes (as demonstrated by the LimitedTxExpansion scenario), are solved cohesively. Details are available in Supplementary Section 5.

Data availability

All input data are referenced in the main text or supplementary information with corresponding citations or repository links. Underlying data for all figures are available via Github at https://github.com/mrziheng/NetZero2050 (ref. 97). Renewable energy resource potential data generated in this study—including installation capacity potential and annual average capacity factors for onshore wind, offshore wind, utility-scale solar PV and rooftop solar PV—are accessible via Github at https://github.com/mrziheng/GlobalRenewableEnergyResource (ref. 98). Linear programming solving files for the GISPO model base scenario are available via Zenodo at https://doi.org/10.5281/zenodo.17618090 (ref. 99). Source data are provided with this paper.

Code availability

All Python scripts for figure generation in the main text, the GISPO model as an open-source Python package and the scenario-specific optimization scripts executing the GISPO model runs for this study are accessible via GitHub at https://github.com/mrziheng/NetZero2050 (ref. 97).

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Competing interests

The authors declare no competing interests.

Additional information

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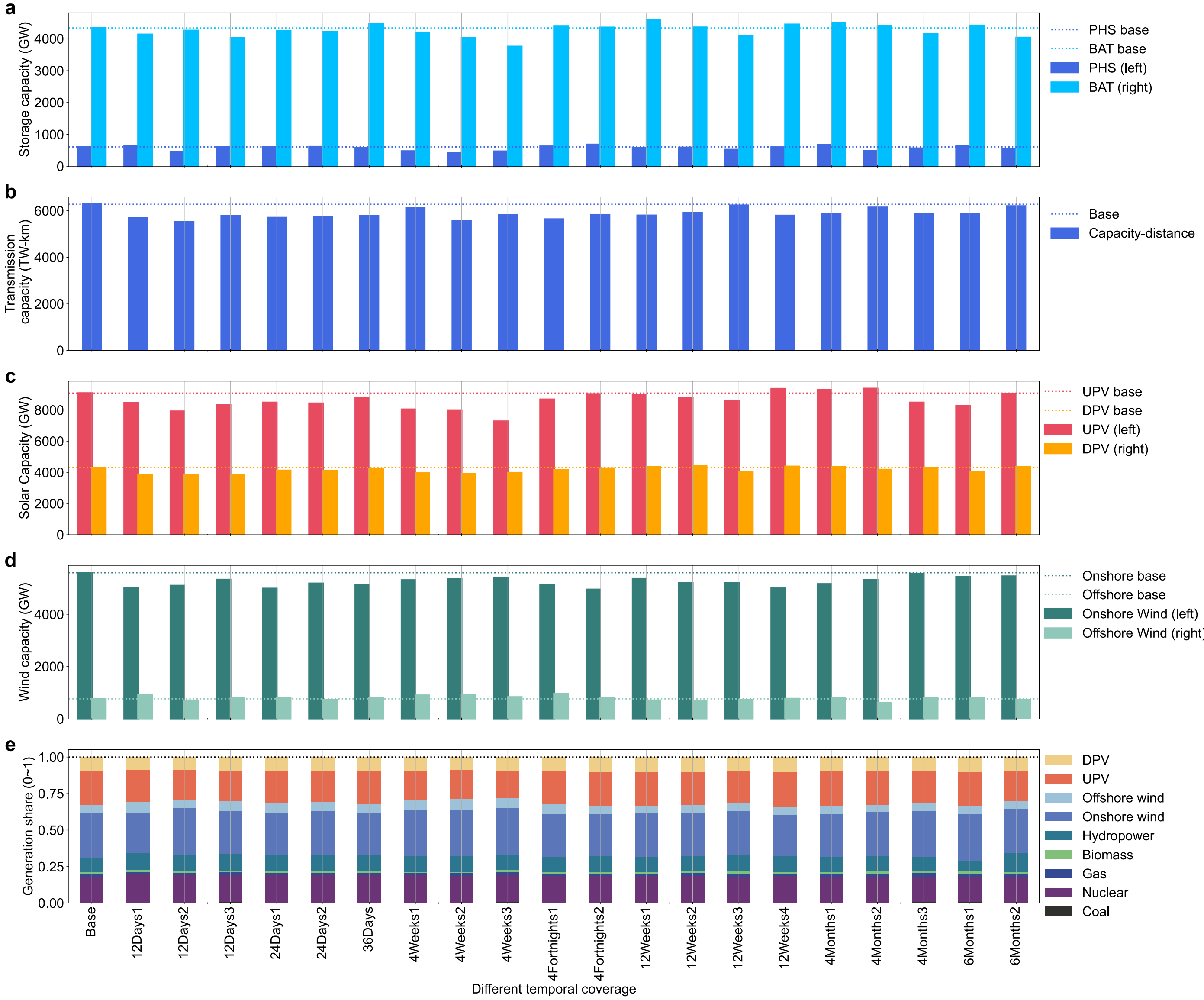

Extended Data Fig. 1 | Sensitivity analysis of the power system expansion model for results in 2050 under different temporal coverage scenarios. (**a**) Capacity (GW) of pumped hydro storage (left) and battery storage (right). (**b**) Transmission capacity (TW-km). (**c**) Capacity (GW) of utility-scale (left) and distributed (right) solar PV. (**d**) Capacity (GW) of onshore wind (left) and offshore wind (right). (**e**) Generation share (0--1) of different generators within the modelling period. UPV: utility-scale photovoltaics; DPV: distributed photovoltaics; PHS: pumped hydro storage; BAT: battery.

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Extended Data Table 1 | Definitions and key findings across scenarios

Category	Scenario	Definition	Key Findings
Baseline	Base	Net-zero power systems for all (58.3 PWh/yr) by 2050.	SCOE: 49 $/MWh; VRE: 70% (19.7 TW); storage: 22.2 TWh; Tx: 6,300 TW-km.
Demand-side	HigherDLS	Higher DLS (3,500 kWh/cap/yr); total +5.5% to 61.5 PWh/yr.	SCOE: ↑<1%; VRE: +5.5% (solar +10.8%, wind +7.8%); storage +7.2%; Tx +6.7%.
	WithoutDLS	Exclude DLS; total demand −6.4%.	SCOE: ↓<1%; DPV: −28.7%, BAT: −9.3%; reduced access in developing regions.
	HigherElec-Demand	Higher electrification (DLS + hydrogen, etc.); total +7.55%.	SCOE: nearly no change; solar: +7%, wind: +11.2%; storage: +6%.
Supply-side	WithoutHP-Expansion	No new hydropower expansion.	Storage: +10% (to 24.6 TWh) compensating for loss of dispatchable hydro.
	SlowerNuclear-Expansion	Nuclear capacity share <3.7% (<1,000 GW); 2024 share 4.1%.	SCOE: ↑<2%; solar: +7.3%, wind: +4.7%; storage: +8.8%.
	LowerVRESupply	Lower land availability for wind and solar.	SCOE: +2%; bioenergy: +7%.
Infrastructure & flexibility	LimitedTx-Expansion	No new cross-border transmission expansion.	SCOE: +5.6%; offshore wind: +92%, storage: +27%; Africa cost: +20%.
	DemandResponse	Grid-level demand response enabled.	SCOE: −6.5%; PHS: −28%, BAT: −16%, Tx: −13%.
	HigherHeat-Electrification	Higher heating electrification with seasonal demand shift.	SCOE: +4.5%; solar: −8.6%, wind: +9.5%; Tx: +7.0%, storage: −8.3%.
	PreRetired-ThermalPower	Early retirement of non-CHP coal and gas by 2050.	solar: −13%, wind: +15.6%; storage: −15.5%, Tx: +17%.
Technology & policy	SlowerTech-Advancement	Delayed cost reductions for renewables and storage.	SCOE: +24%; UPV: −18.5%, BAT: −60%; offshore wind: +90%, BECCS: +30%.
	FreeTrade	Investment costs at the lowest regional values (China).	SCOE: −12% ($345B/yr saved); Africa: cost −20%; benefits developed regions.
	WithoutEmisCap	No carbon emissions constraints.	SCOE: −11.5%; global electricity emissions: 11 Gt/yr.
Combined	DemandingDLS	Combines HigherDLS, LowerVRESupply, LimitedTxExpansion, SlowerTechAdvancement.	SCOE: +31%; NETs: 3.5× base.

If not specified (for example, WithoutEmisCap scenario), net-zero carbon emissions are included in each scenario. SCOE: system cost of electricity; VRE: variable renewable energy; DLS: decent living standards; PV: photovoltaics; UPV: utility-scale PV; DPV: distributed PV; PHS: pumped hydro storage; BAT: battery; Tx: transmission; NETs: net-emissions technologies.